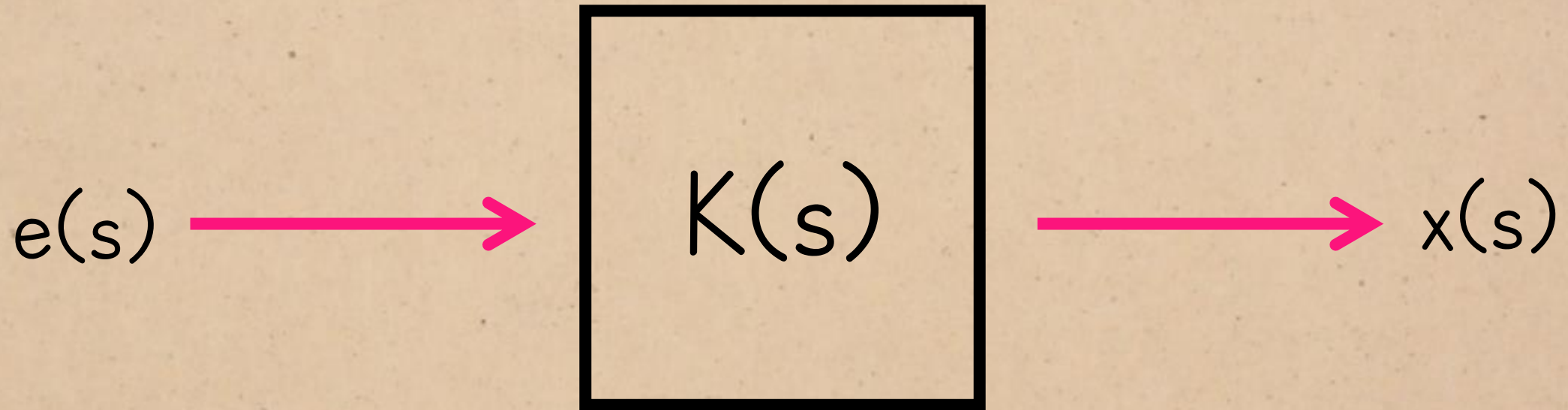


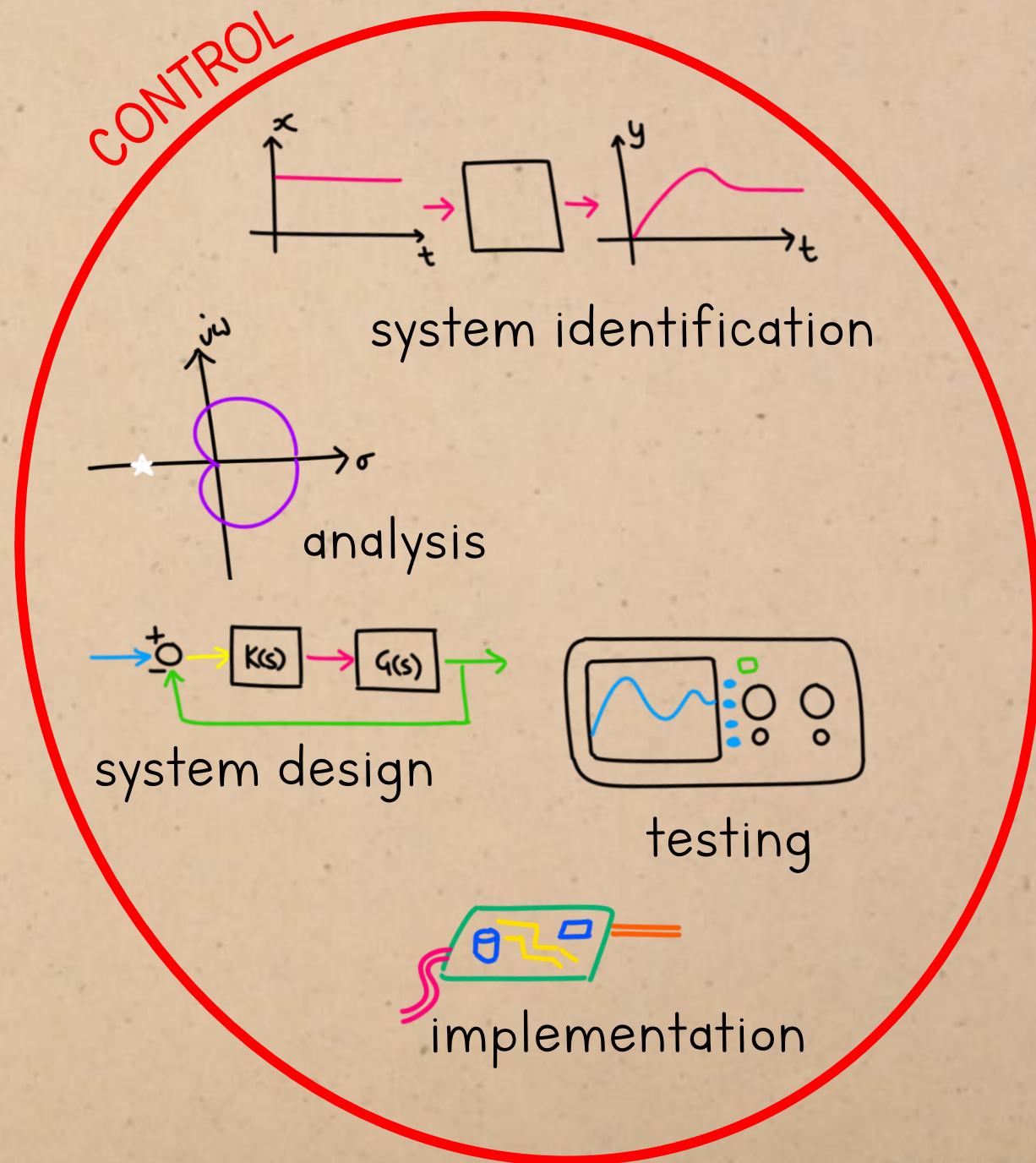


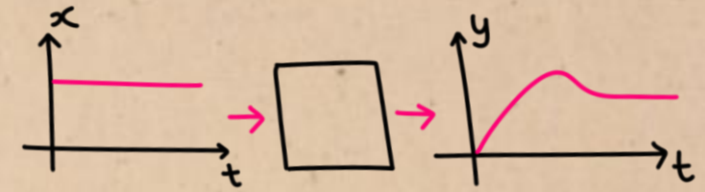
CHAPTER 8:

COMPENSATION

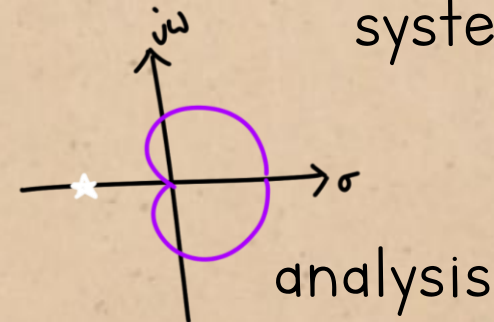
THE BASICS

In this chapter, we're finally getting to the **control** part of control engineering, but we often use the word "control" to mean the whole process...

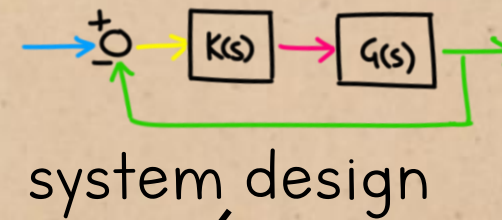




system identification



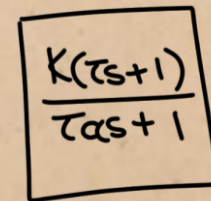
analysis



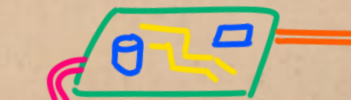
system design



testing



compensation



implementation

So I'm going to use "**compensation**" to refer to the specific task of devising a control law.

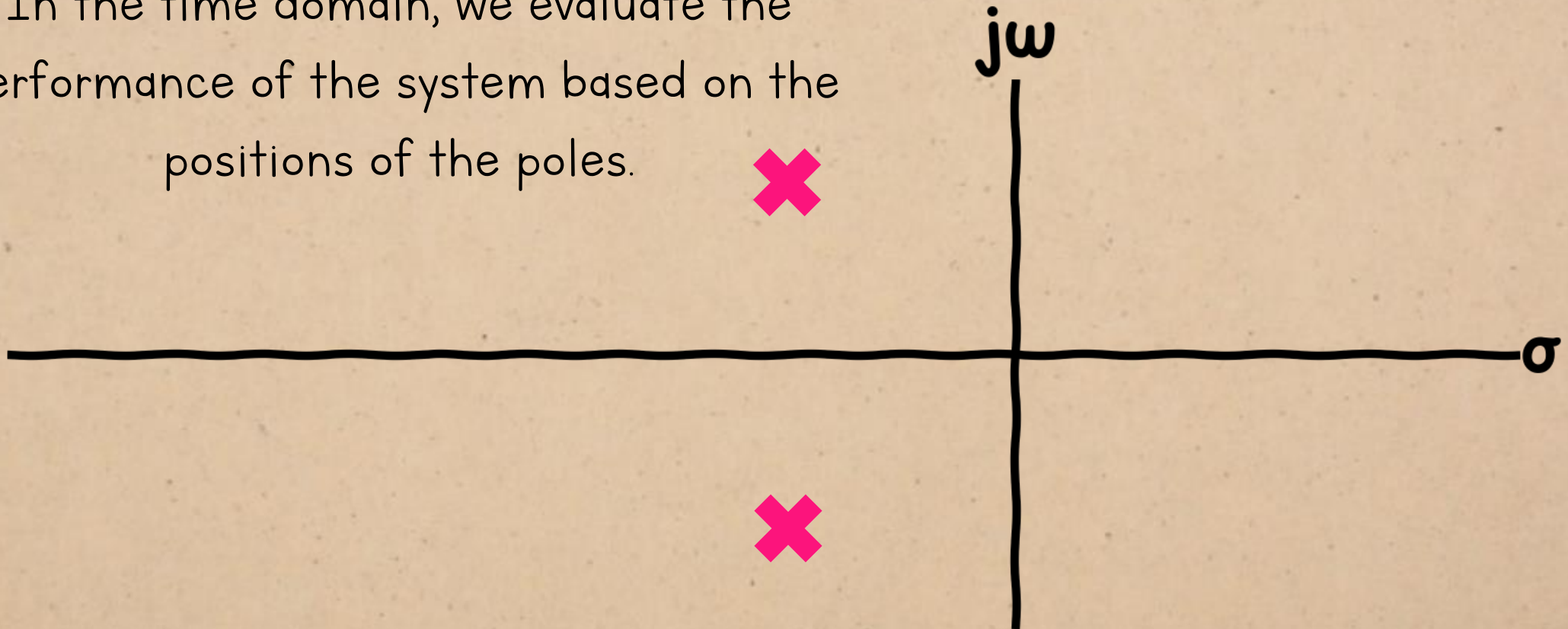
A **compensator** is another word for a controller.

In this lecture, we're going to look at the basic principles of compensation.

You already know more about it than you might expect, in that you have a clear idea of the end goal: you know what a **good** system looks like, and can even define one in two different domains.

TIME DOMAIN

In the time domain, we evaluate the performance of the system based on the positions of the poles.

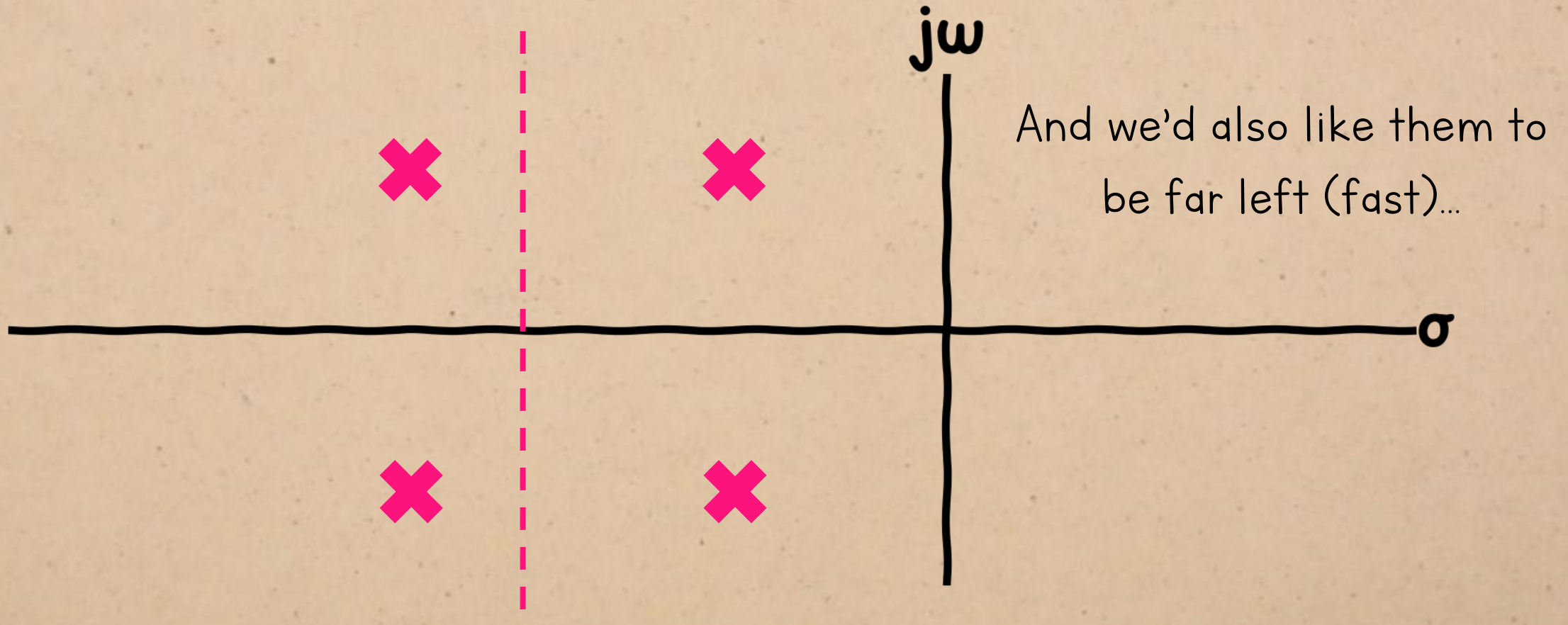


TIME DOMAIN

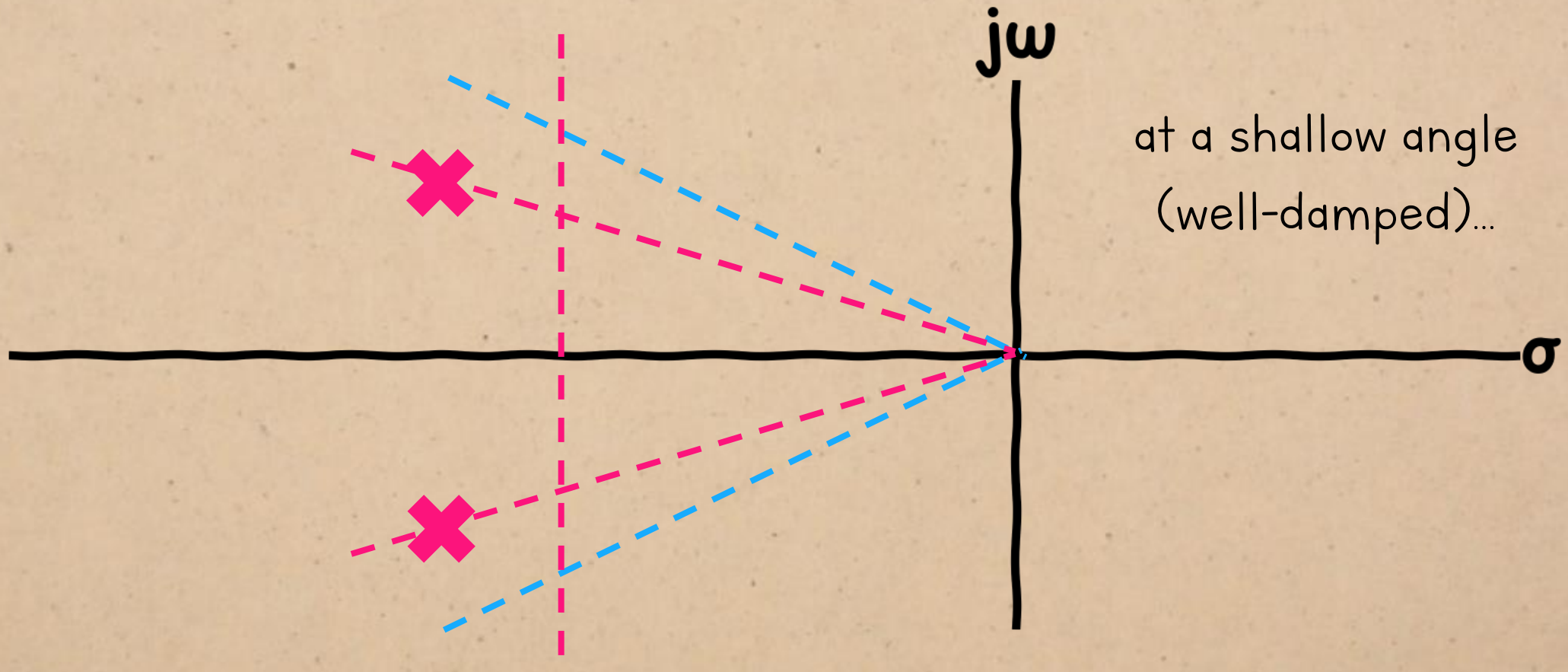
They need to be in the left-half plane (stable)



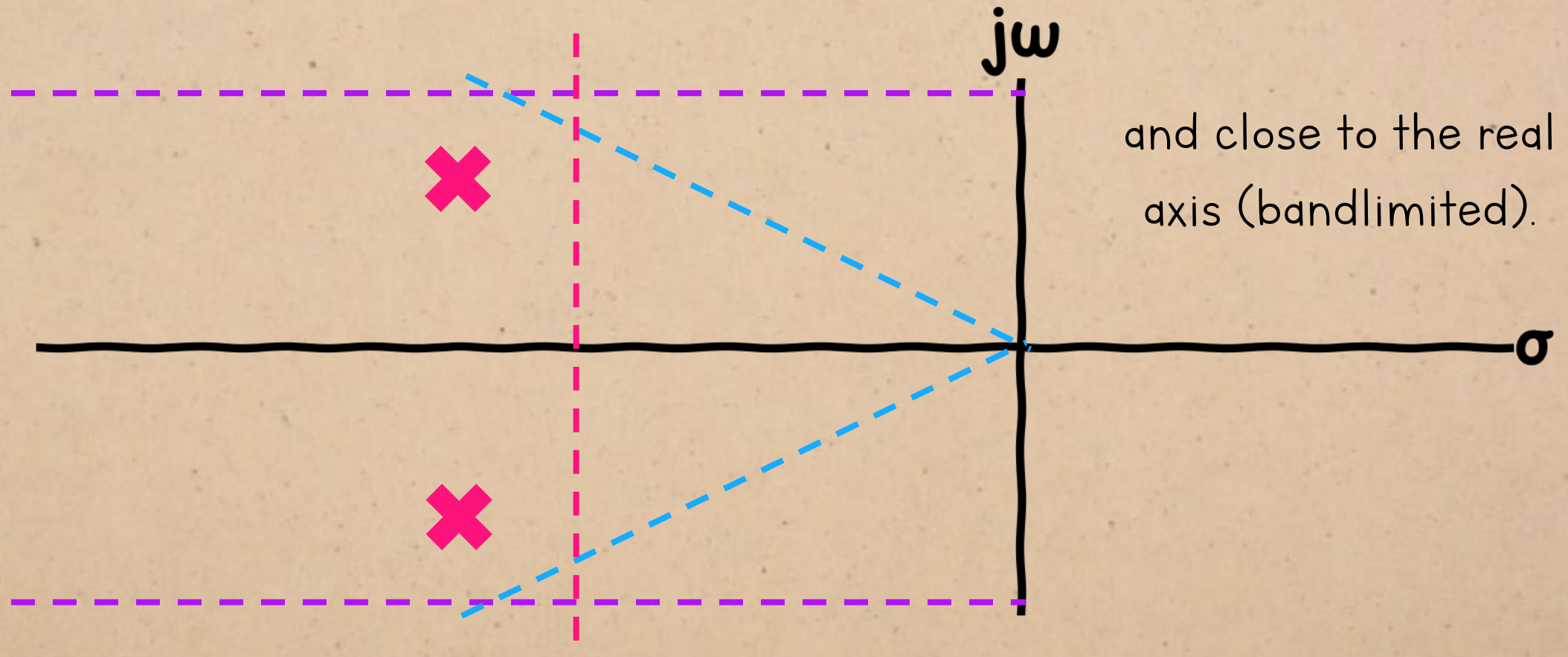
TIME DOMAIN



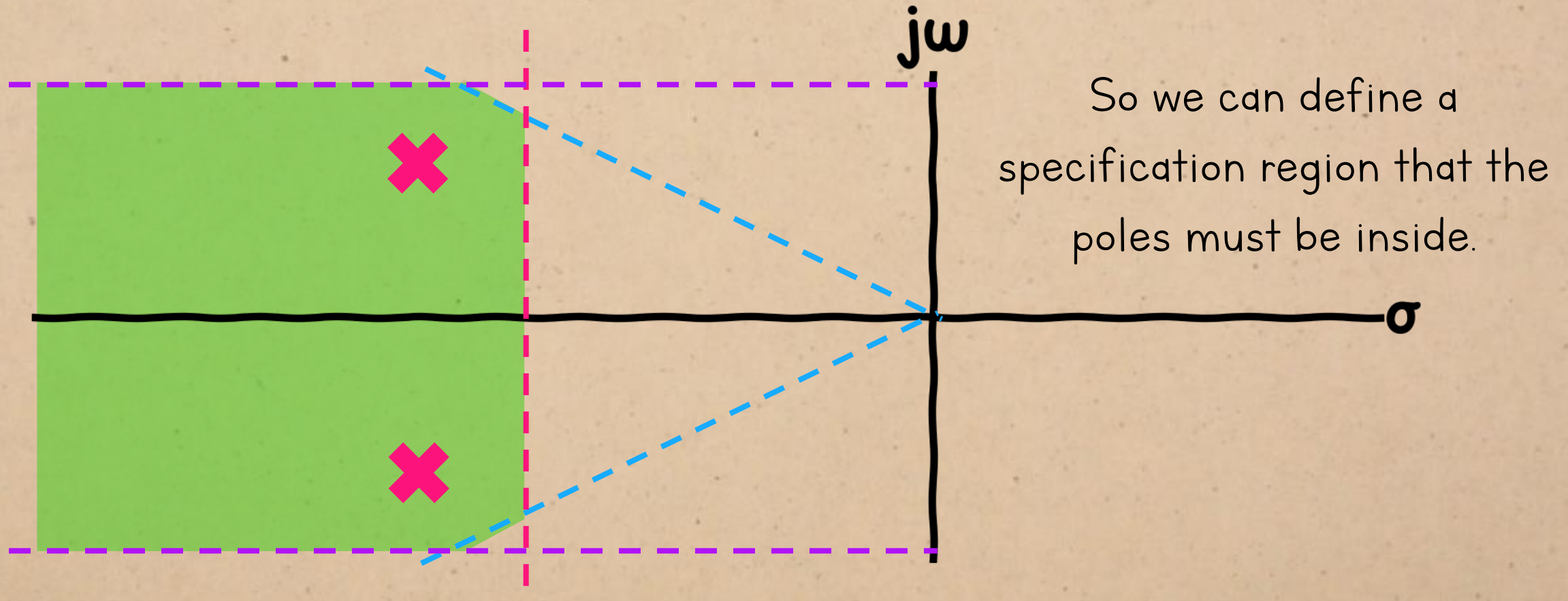
TIME DOMAIN



TIME DOMAIN

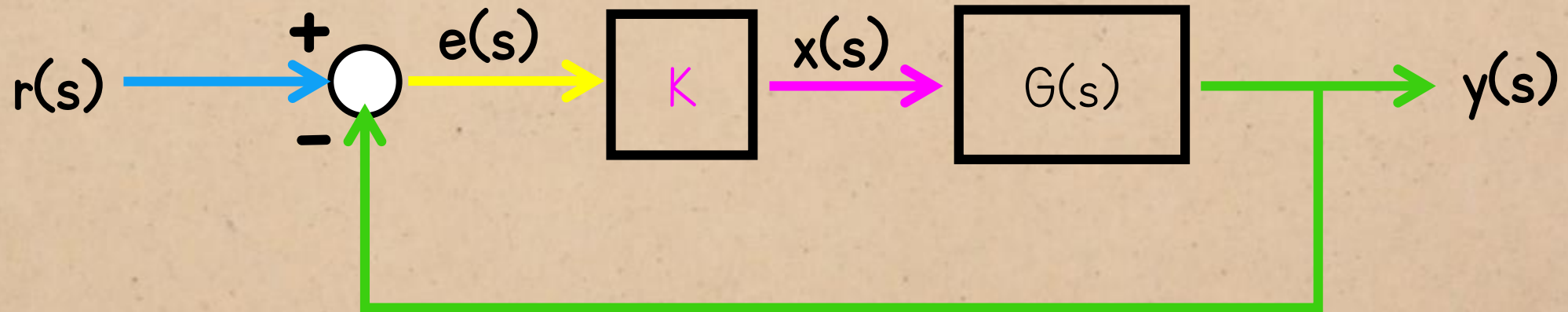


TIME DOMAIN



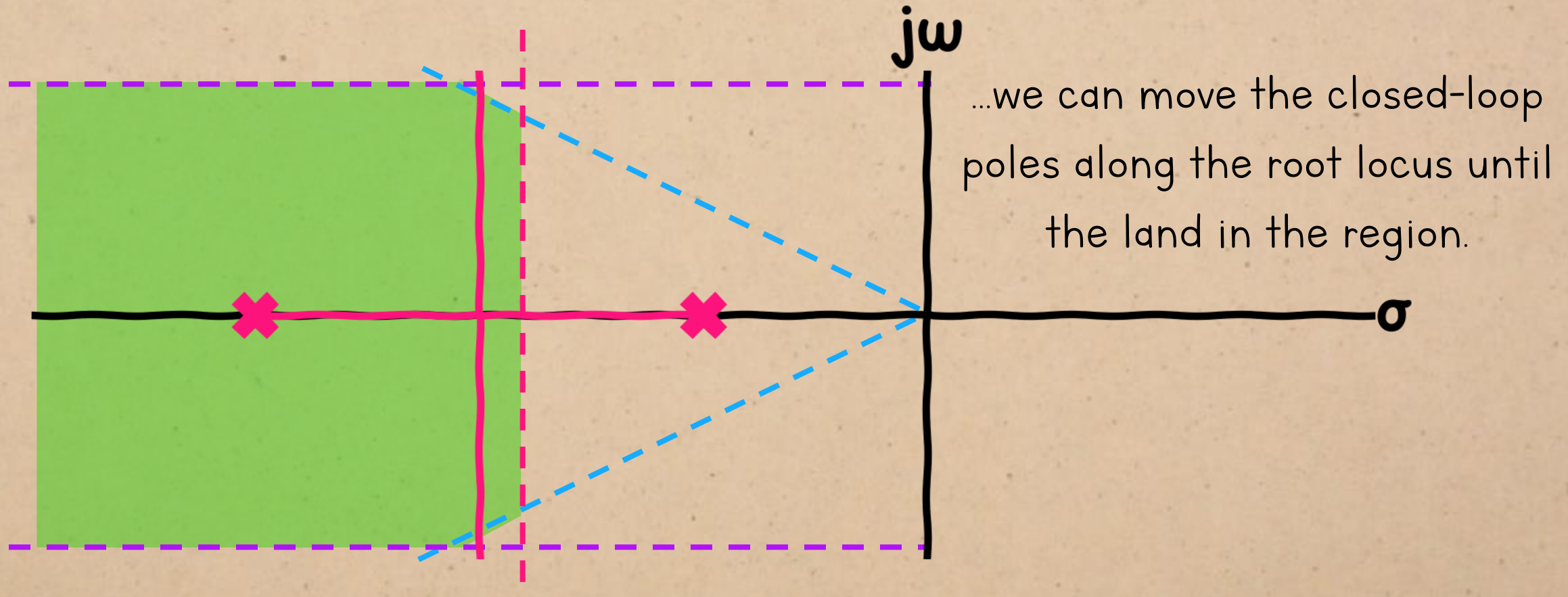
TIME DOMAIN

The simplest way to get the poles where we want them to be is **proportional control**.

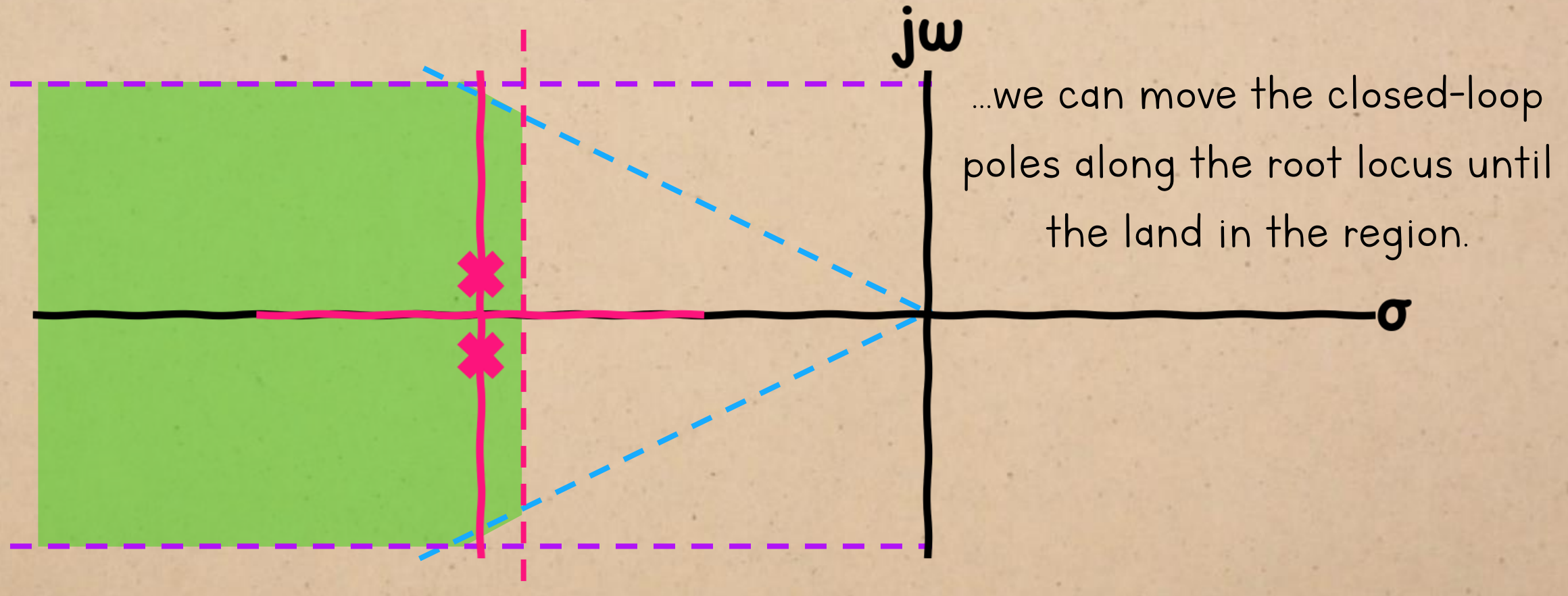


By changing the gain of the open-loop transfer function...

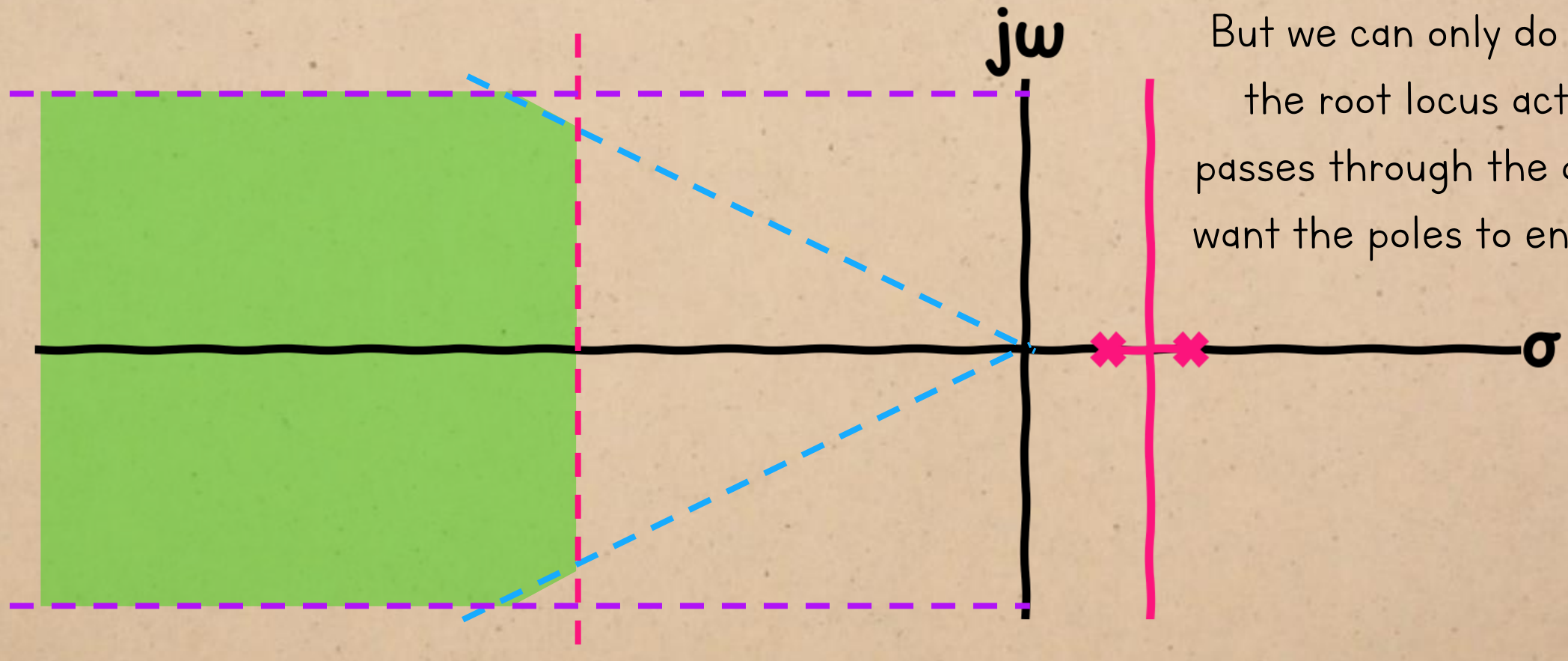
TIME DOMAIN



TIME DOMAIN



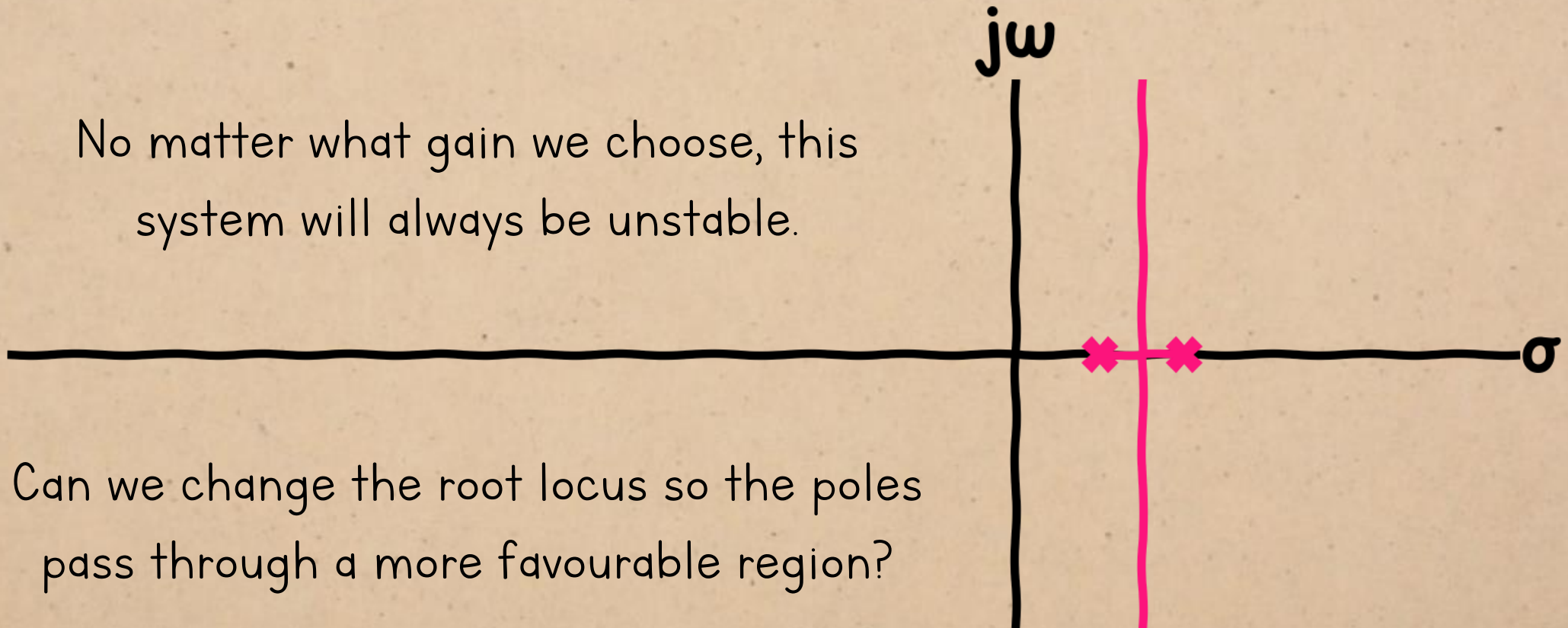
TIME DOMAIN



But we can only do that if
the root locus actually
passes through the area we
want the poles to end up in.

TIME DOMAIN

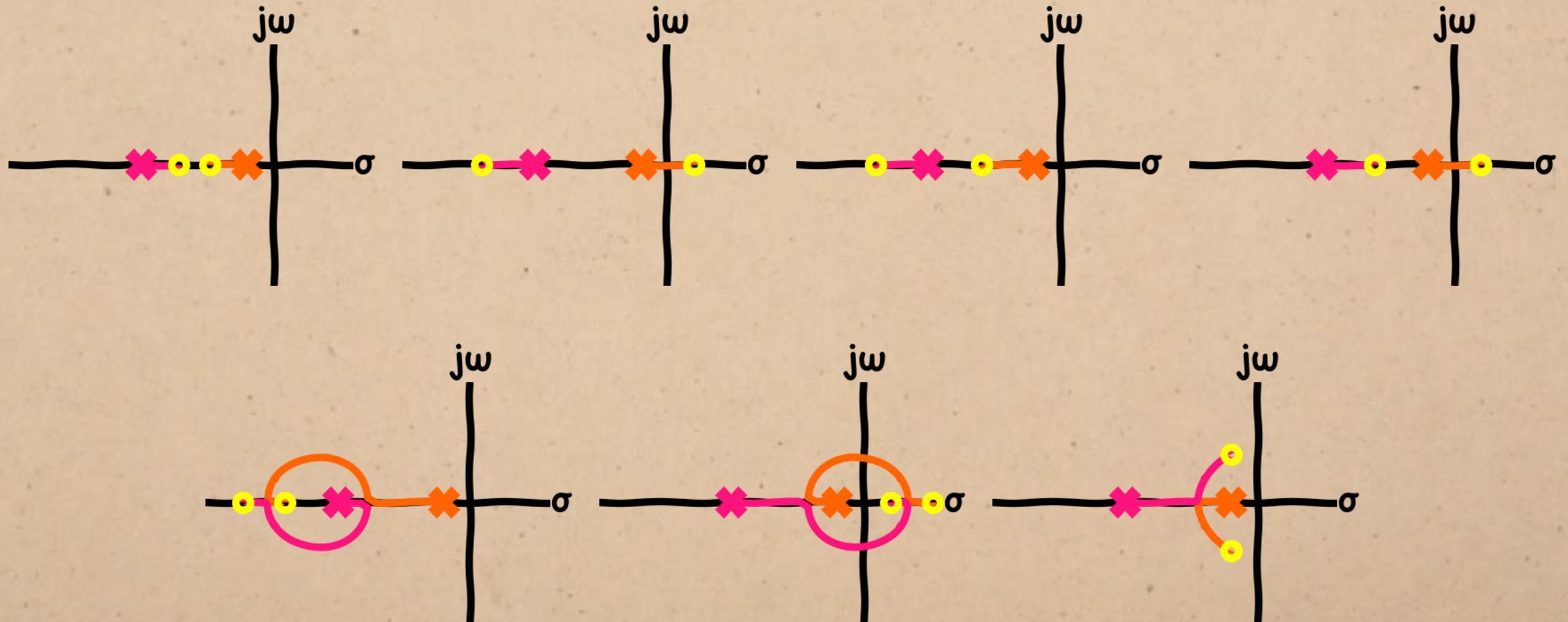
No matter what gain we choose, this system will always be unstable.



Can we change the root locus so the poles pass through a more favourable region?

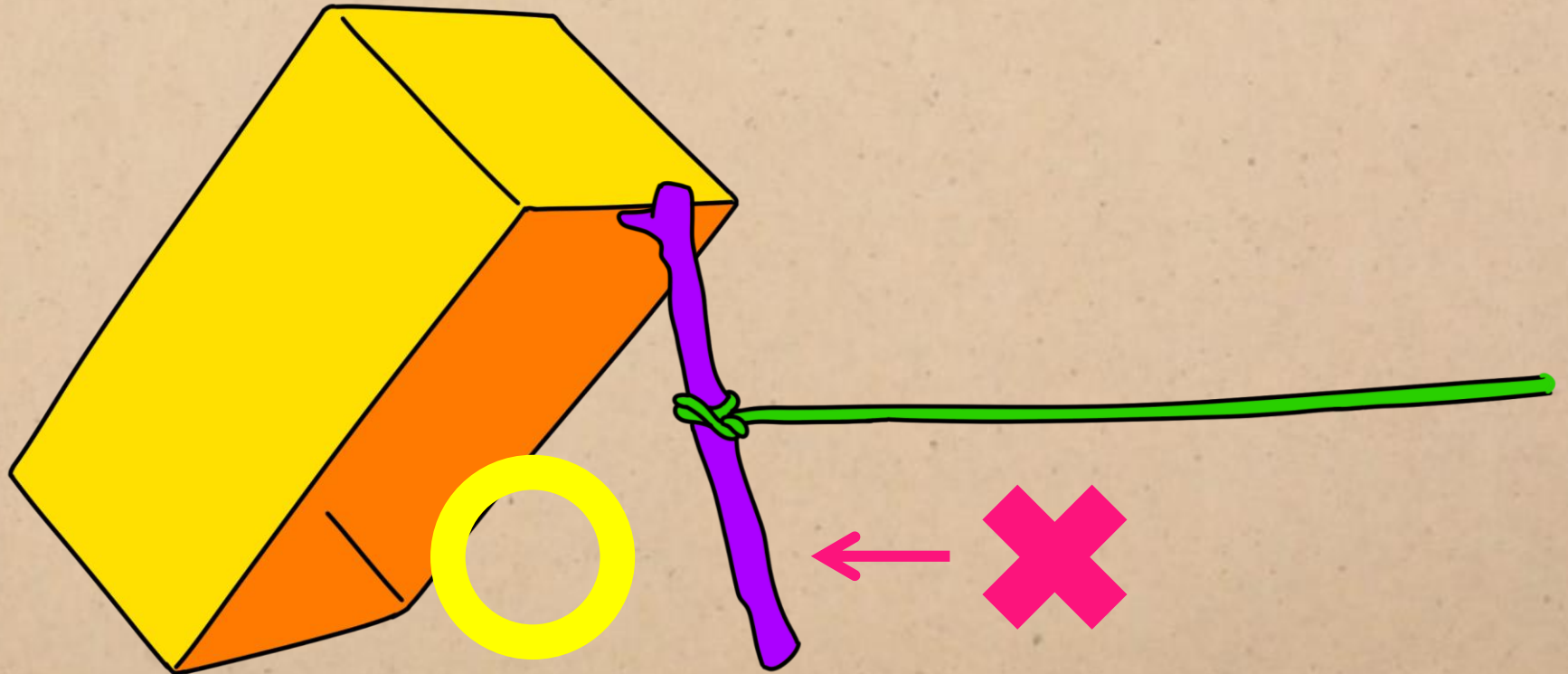
TIME DOMAIN

We understand how the closed-loop poles move in relation to the open-loop poles and zeros.

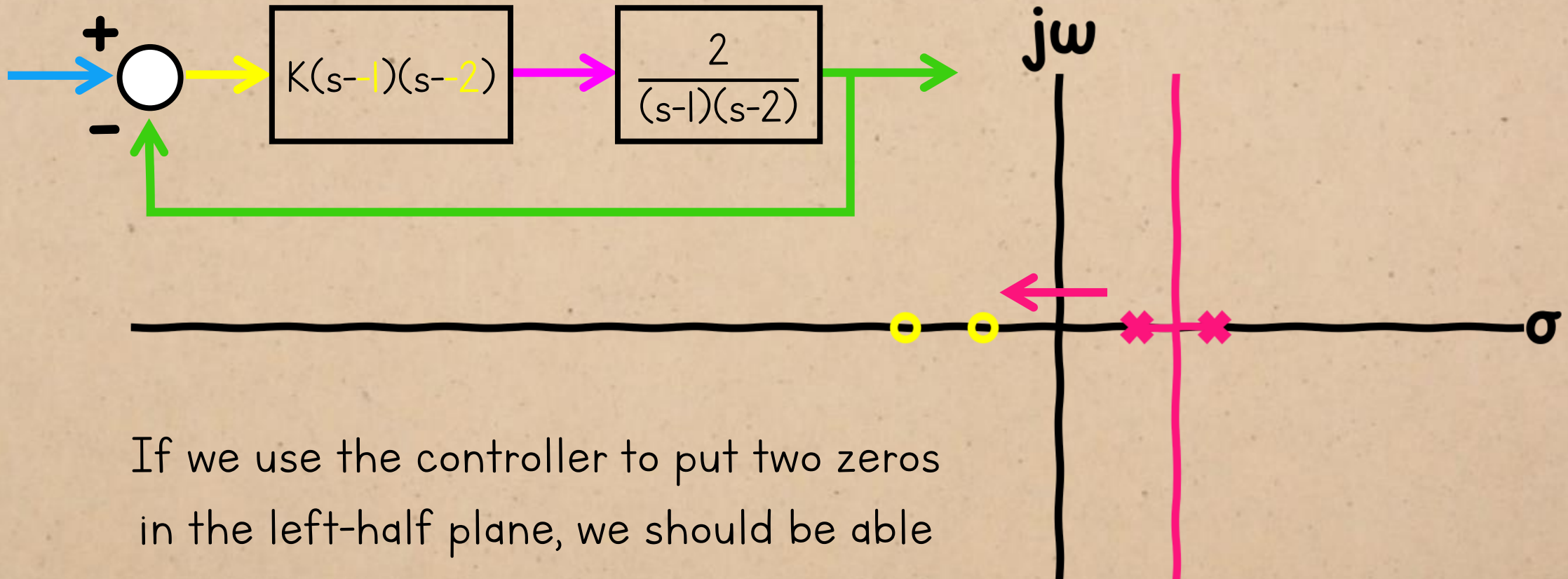


TIME DOMAIN

We know that if there's a zero somewhere, one of the poles will travel towards it.

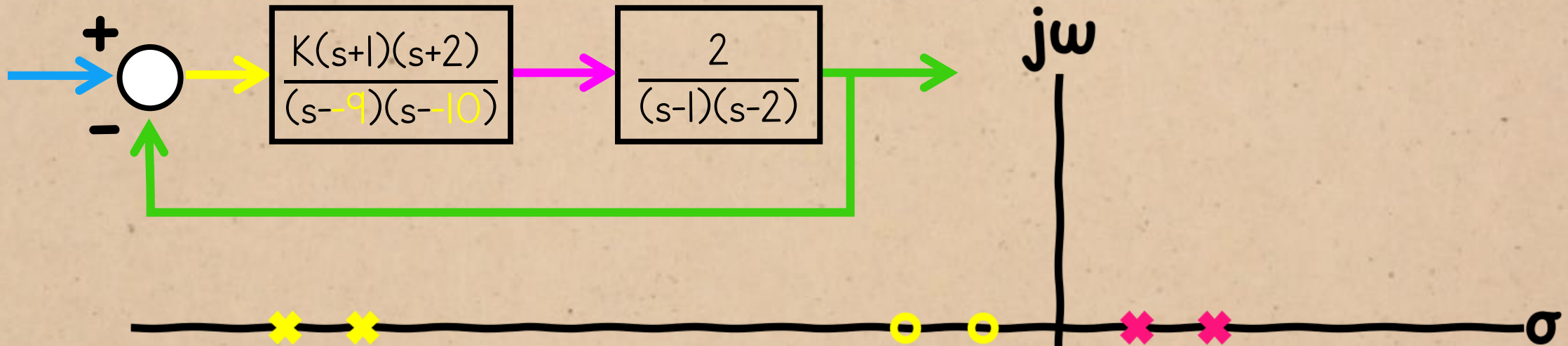


TIME DOMAIN



If we use the controller to put two zeros in the left-half plane, we should be able to attract the poles towards that side.

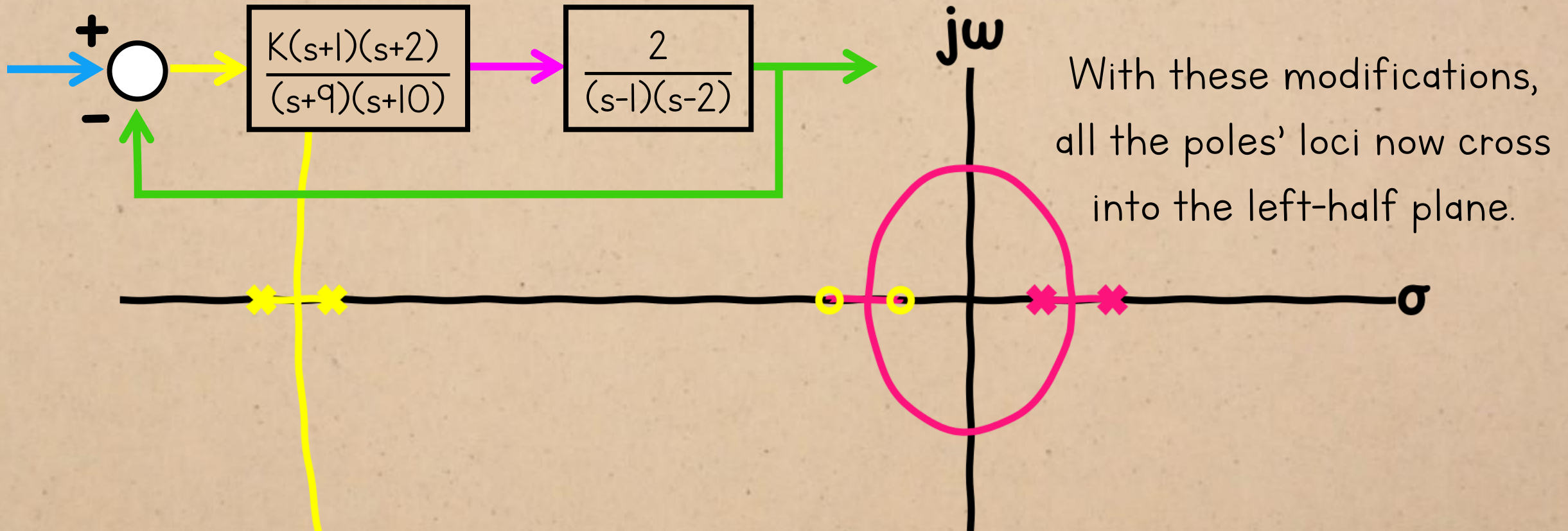
TIME DOMAIN



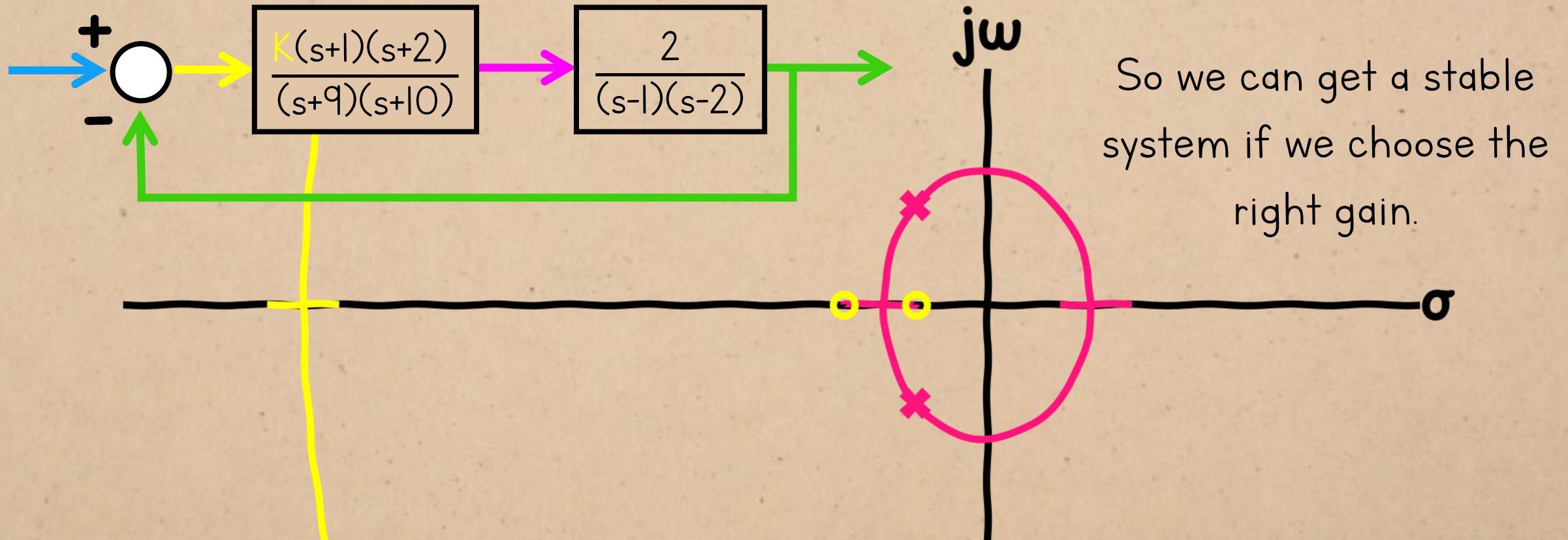
We also need to add a non-dominant pole for each of the zeros, so the controller will have a **proper** transfer function.

(I'll give you more information about how to actually choose these values in later lectures, but for now, I just want to demonstrate the general mechanisms we use to manipulate the system's behaviour)

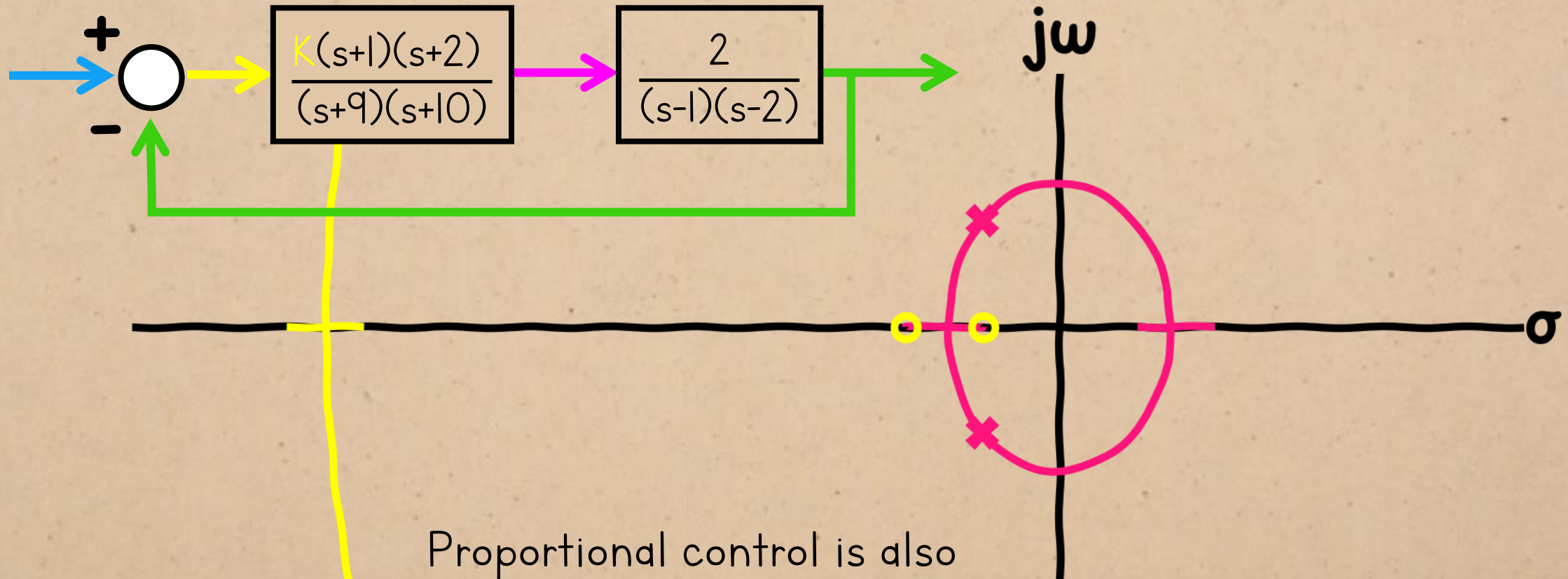
TIME DOMAIN



TIME DOMAIN

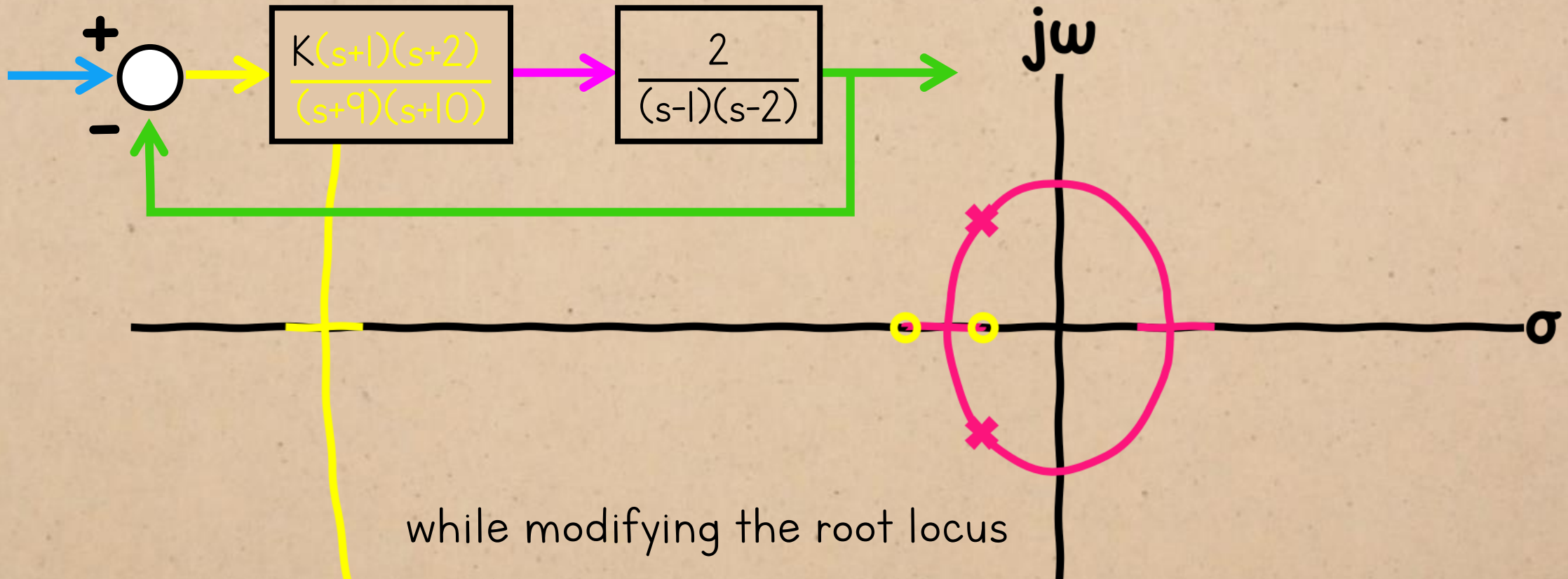


TIME DOMAIN



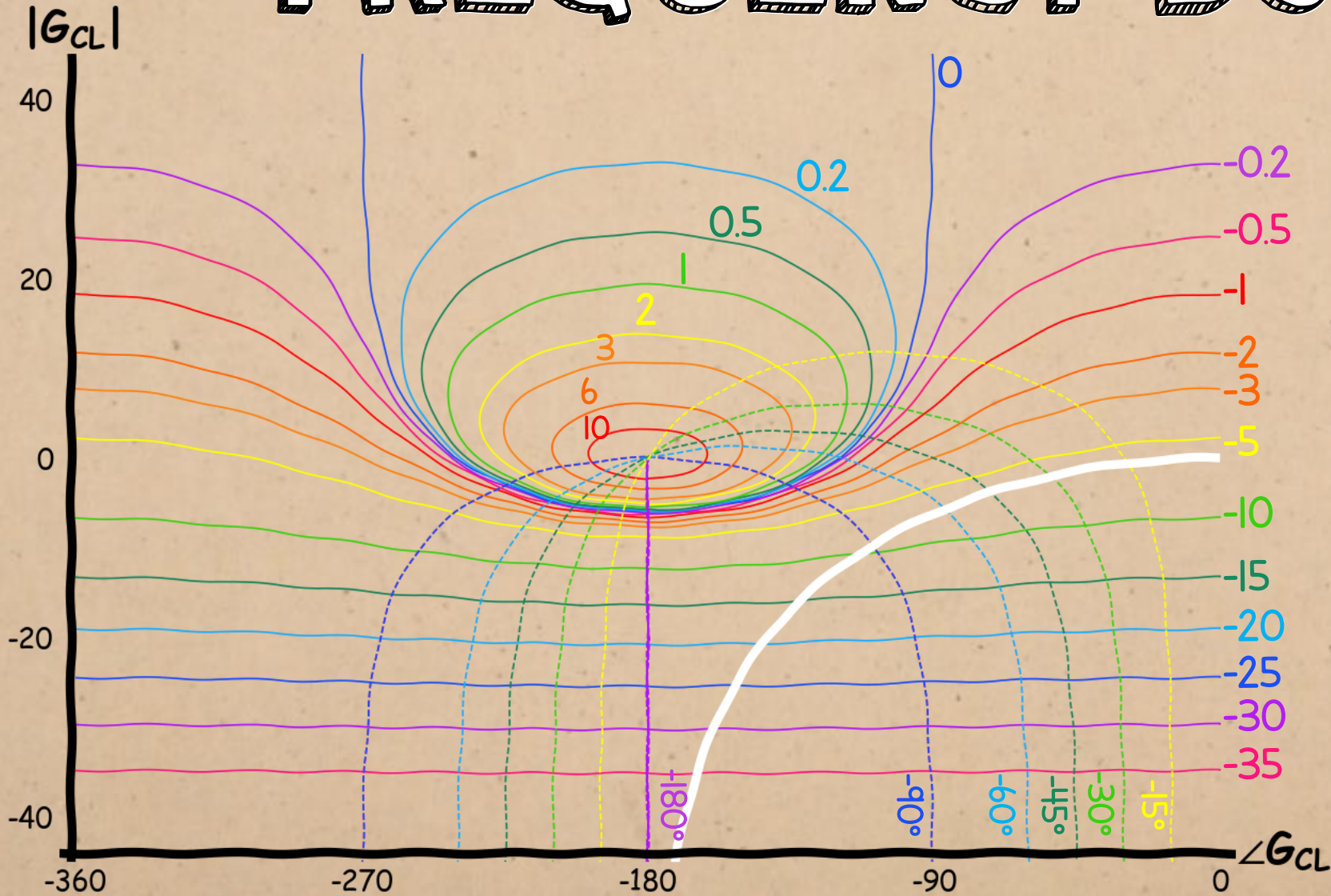
Proportional control is also called **gain compensation**,

TIME DOMAIN



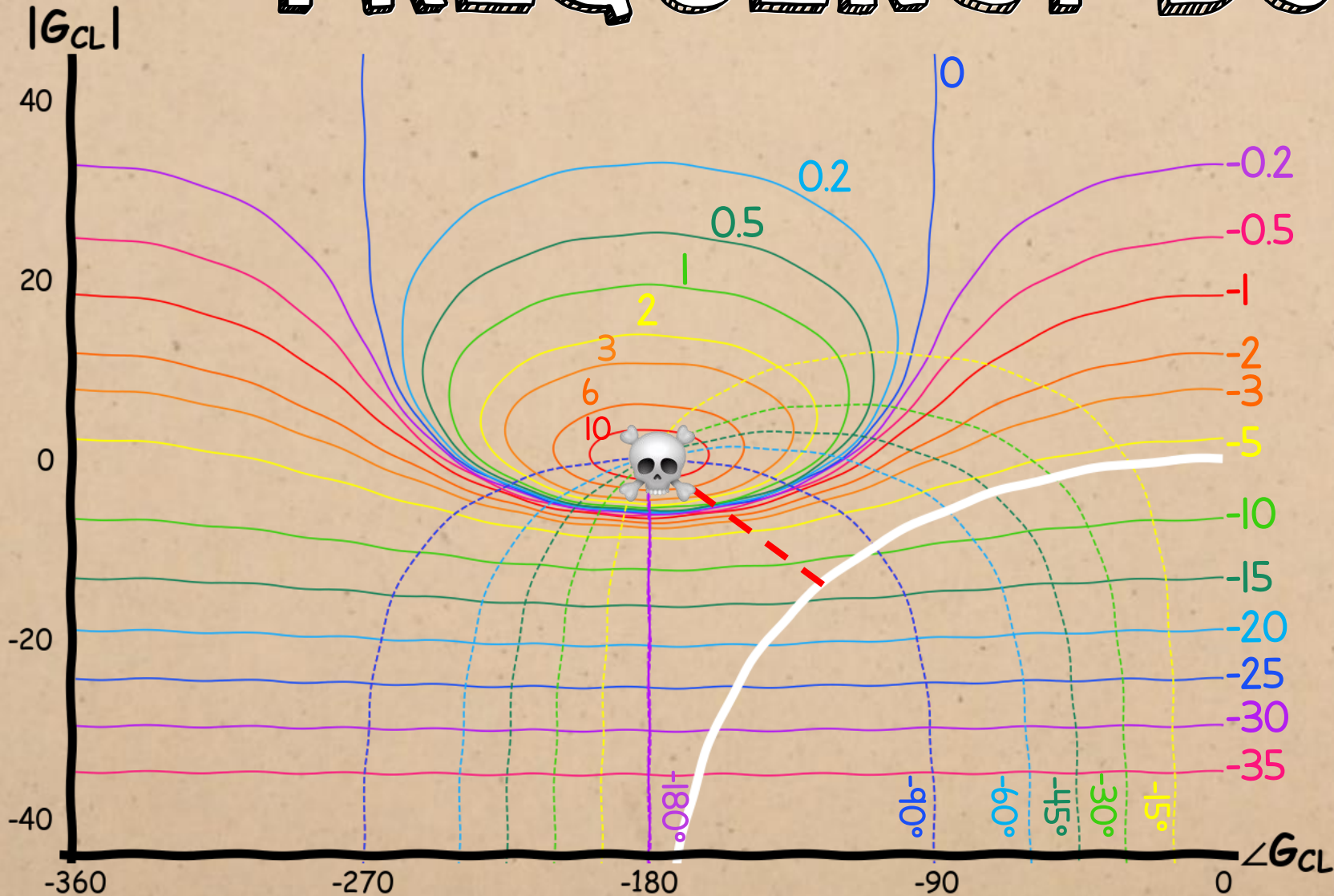
while modifying the root locus
by adding extra zeros and poles
is **gain and phase compensation**.

FREQUENCY DOMAIN



We can also define what a good system looks like in the frequency domain.

FREQUENCY DOMAIN



We want the response to be far away from any point representing -1

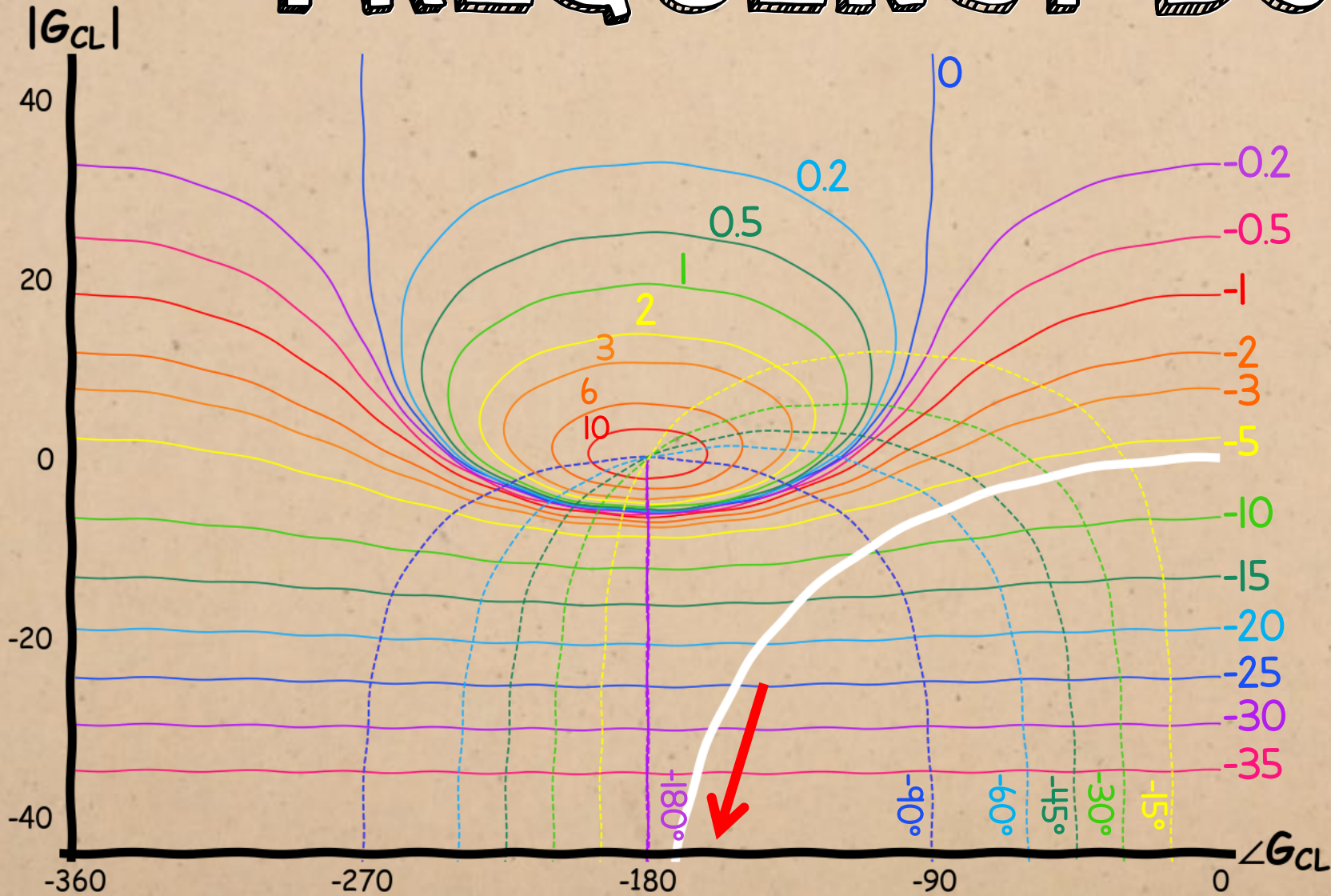
i.e. where a 0 dB gain coincides with an angle that is an odd multiple of 180° (closed-loop stable)

FREQUENCY DOMAIN



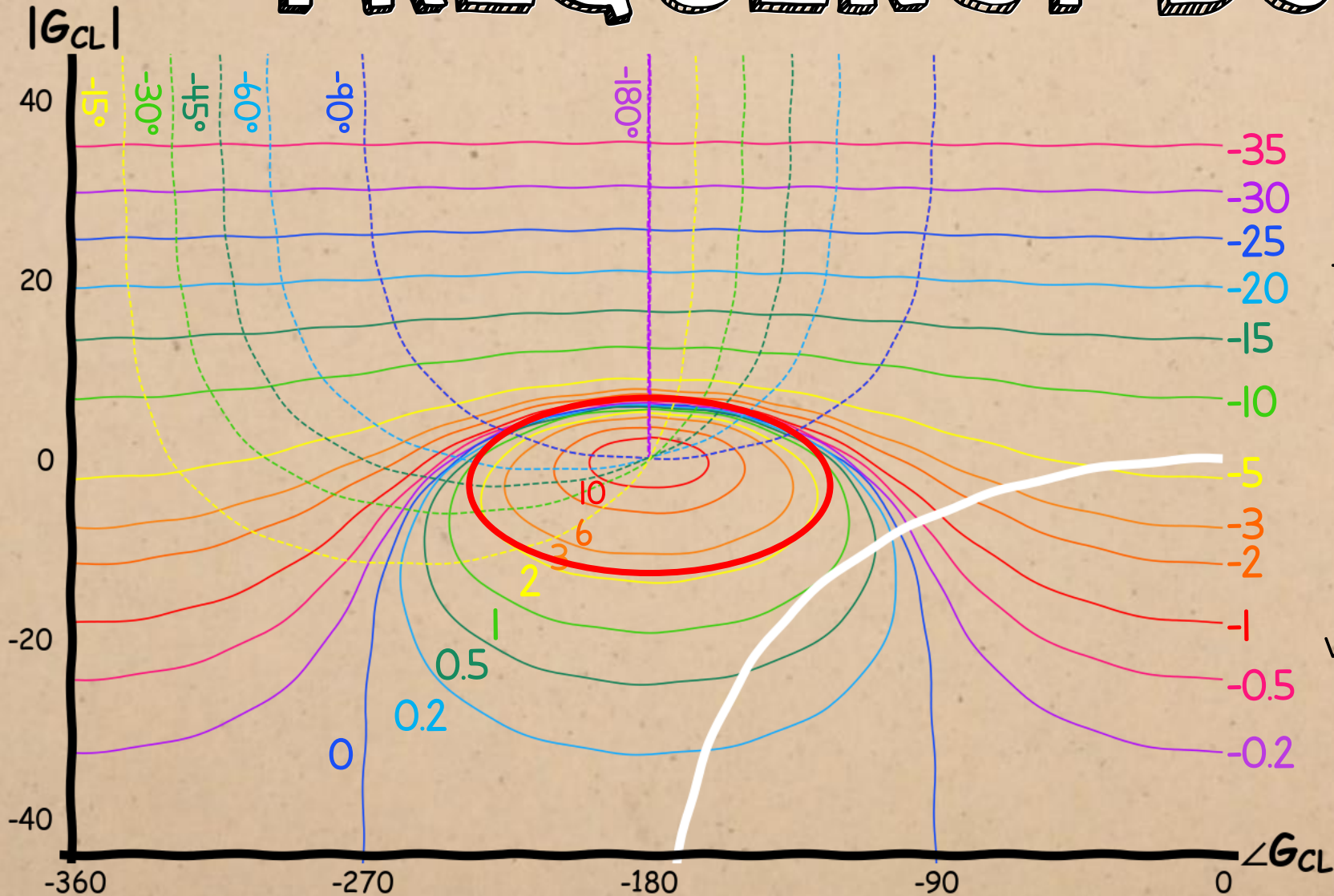
High gain at **low** frequencies is good, because it typically improves setpoint tracking and disturbance rejection...

FREQUENCY DOMAIN



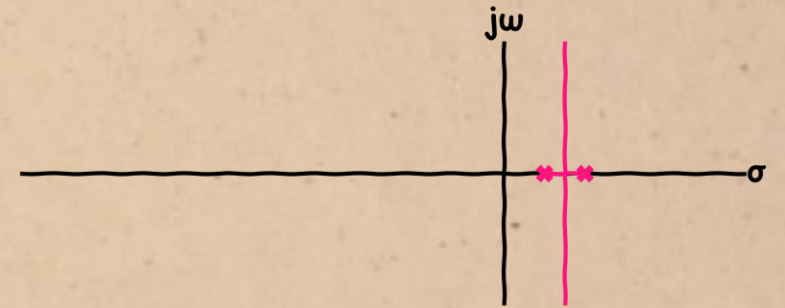
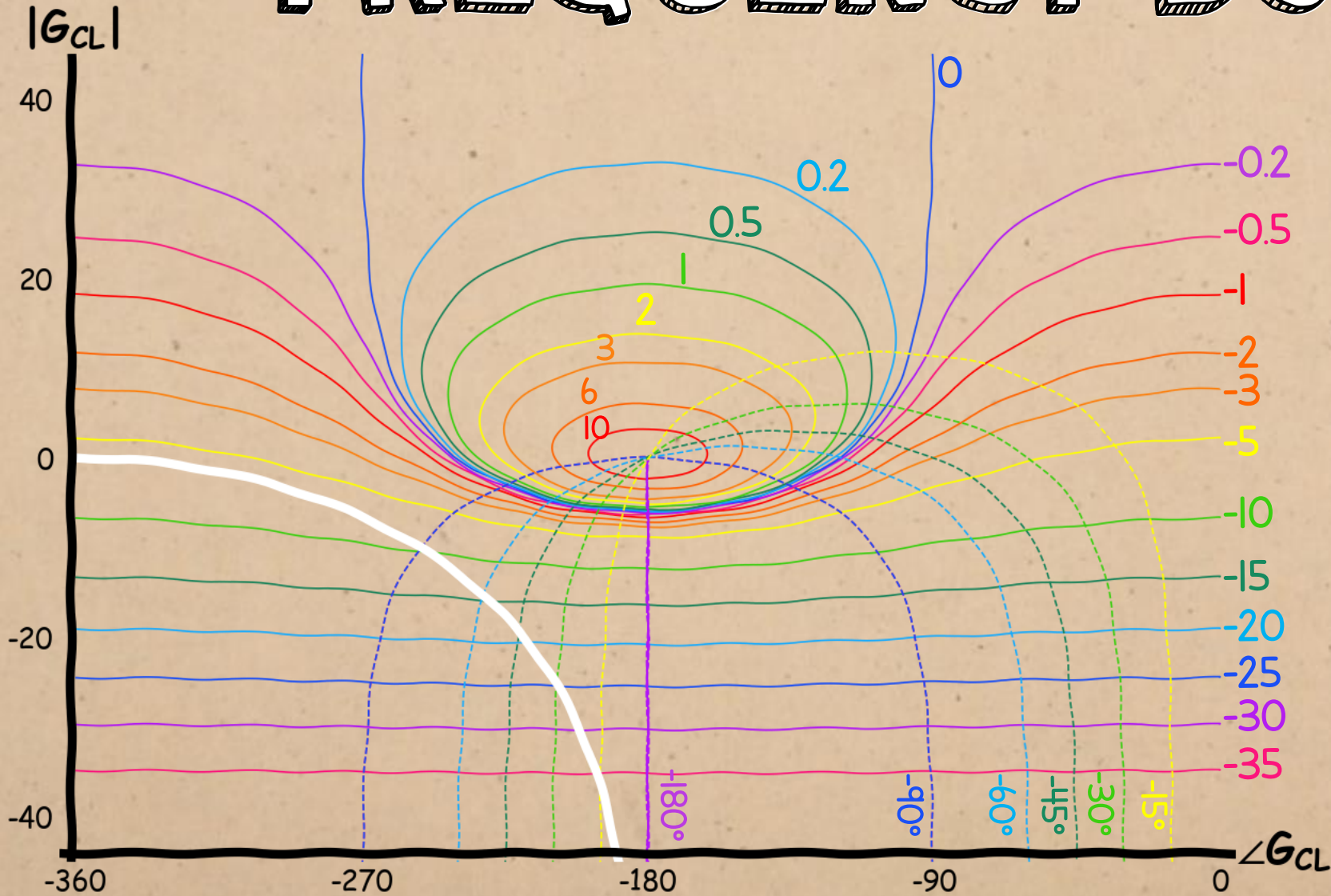
But we want **low** gain at **high** frequencies, so the system attenuates noise.

FREQUENCY DOMAIN



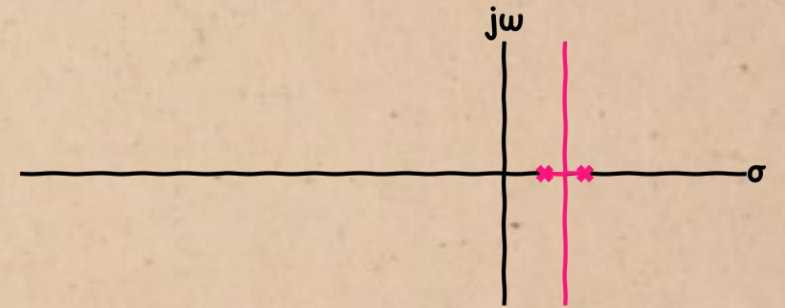
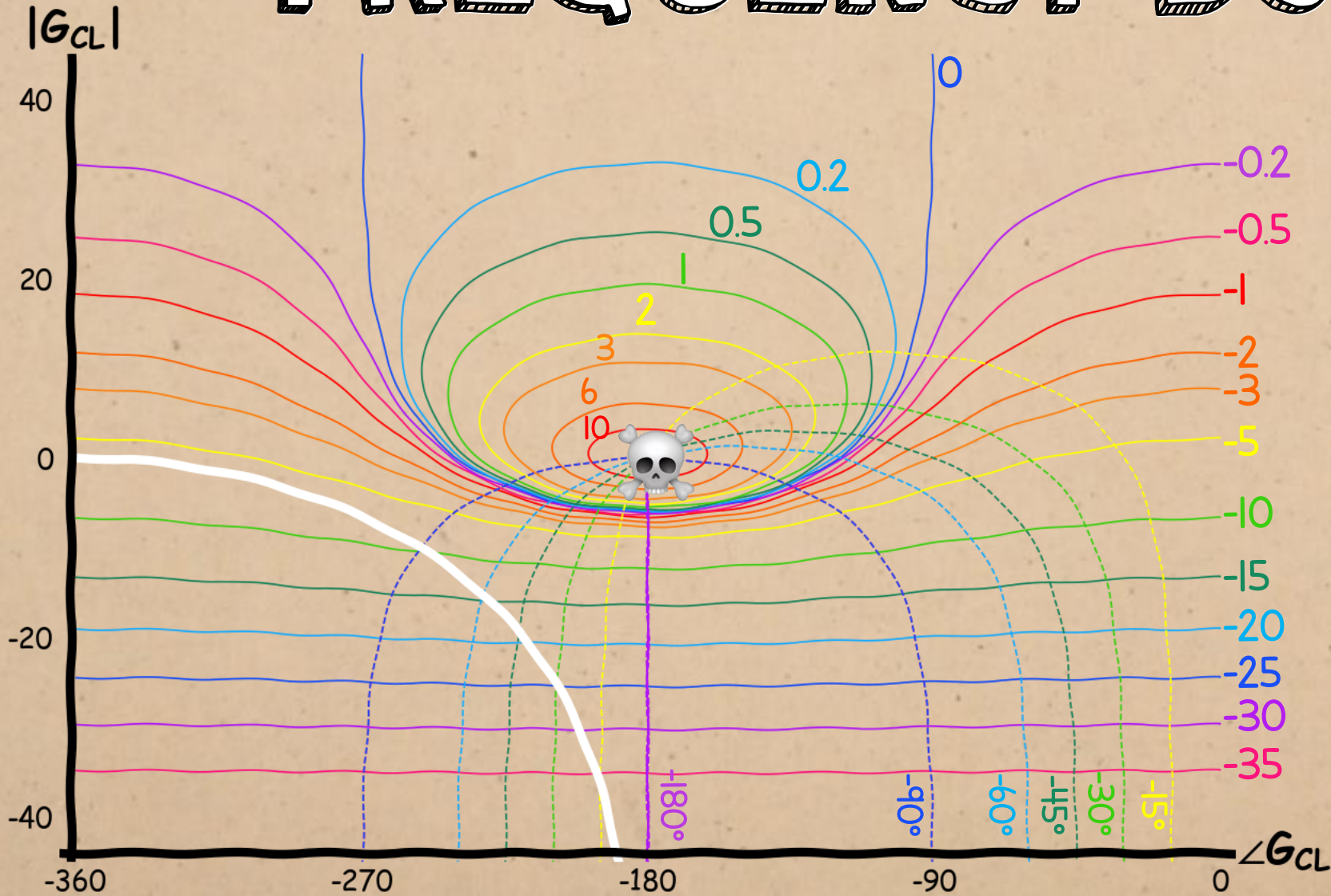
The inverse Nichols chart shows that a response with this shape also avoids large sensitivity function magnitudes, so the closed-loop system is **robust** to variation in the open-loop model.

FREQUENCY DOMAIN



This is the unstable system from the previous time-domain example.

FREQUENCY DOMAIN



If it's unstable, why doesn't it cross -1?

FREQUENCY DOMAIN

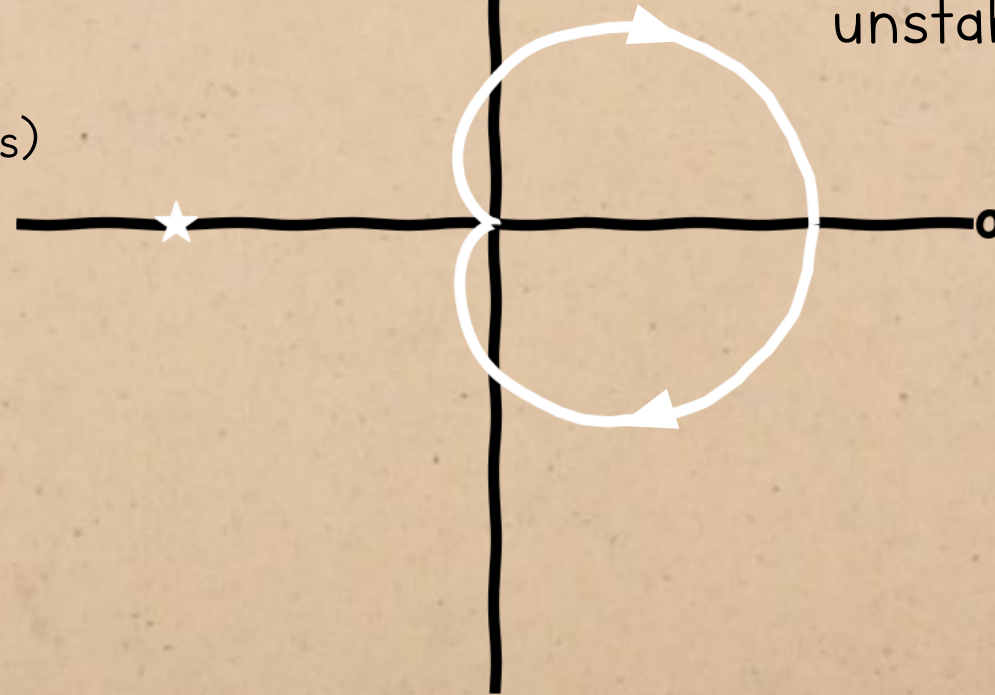
clockwise encirclements of -1

$$N = Z - P$$

unstable poles $G(s)$

$j\omega$

right-half zeros of $1 + G(s)$



Remember how the Nyquist stability criterion applies to systems that already have unstable poles in open-loop...

FREQUENCY DOMAIN

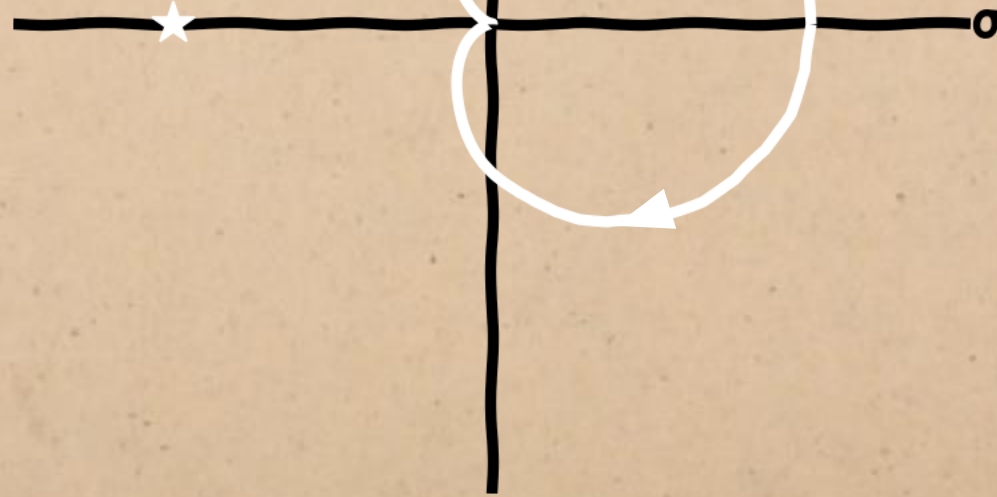
clockwise encirclements of -1

$$N = Z - P$$

unstable poles $G(s)$

$j\omega$

= unstable closed-loop poles



FREQUENCY DOMAIN

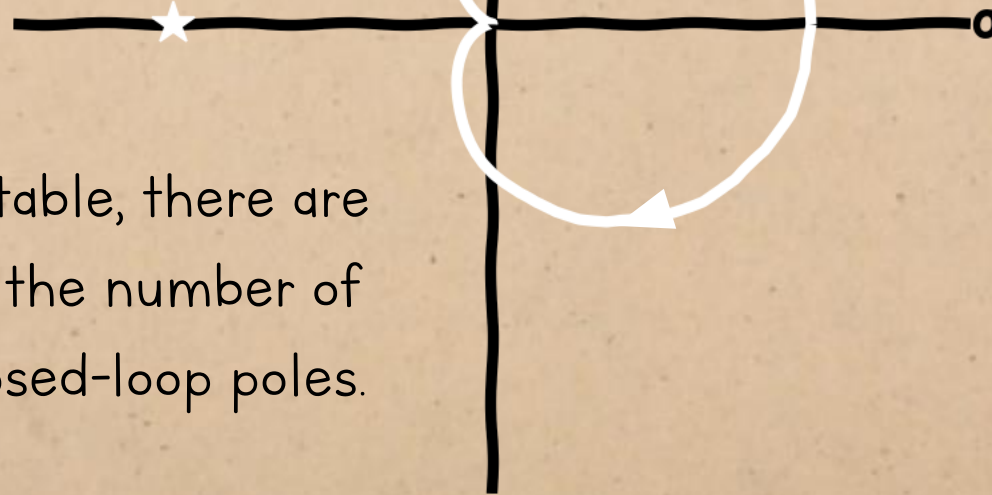
clockwise encirclements of -1

$$N = Z - O$$

unstable poles $G(s)$

$j\omega$

= unstable closed-loop poles



If the system is open-loop stable, there are no right-half plane poles, so the number of encirclements = unstable closed-loop poles.

FREQUENCY DOMAIN

= unstable closed-loop poles

N

=

Z

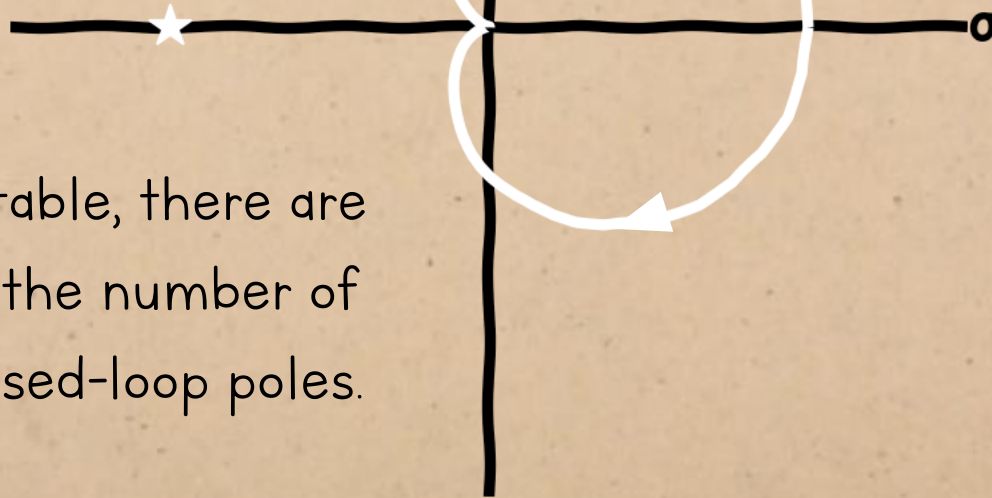
-

O

unstable poles $G(s)$

$j\omega$

= unstable closed-loop poles



If the system is open-loop stable, there are no right-half plane poles, so the number of encirclements = unstable closed-loop poles.

FREQUENCY DOMAIN

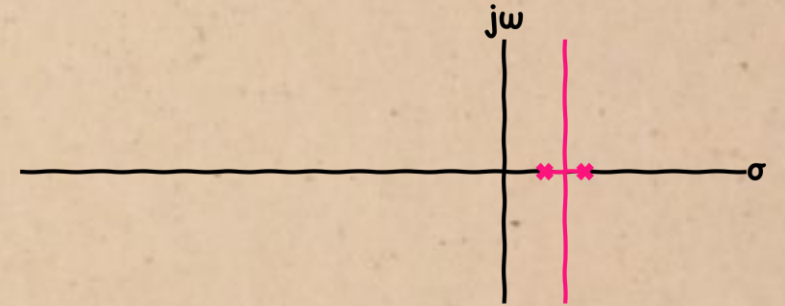
clockwise encirclements of -1

$$N = Z - 2$$

= unstable closed-loop poles

unstable poles $G(s)$

$j\omega$



But this system has two unstable poles in open loop,

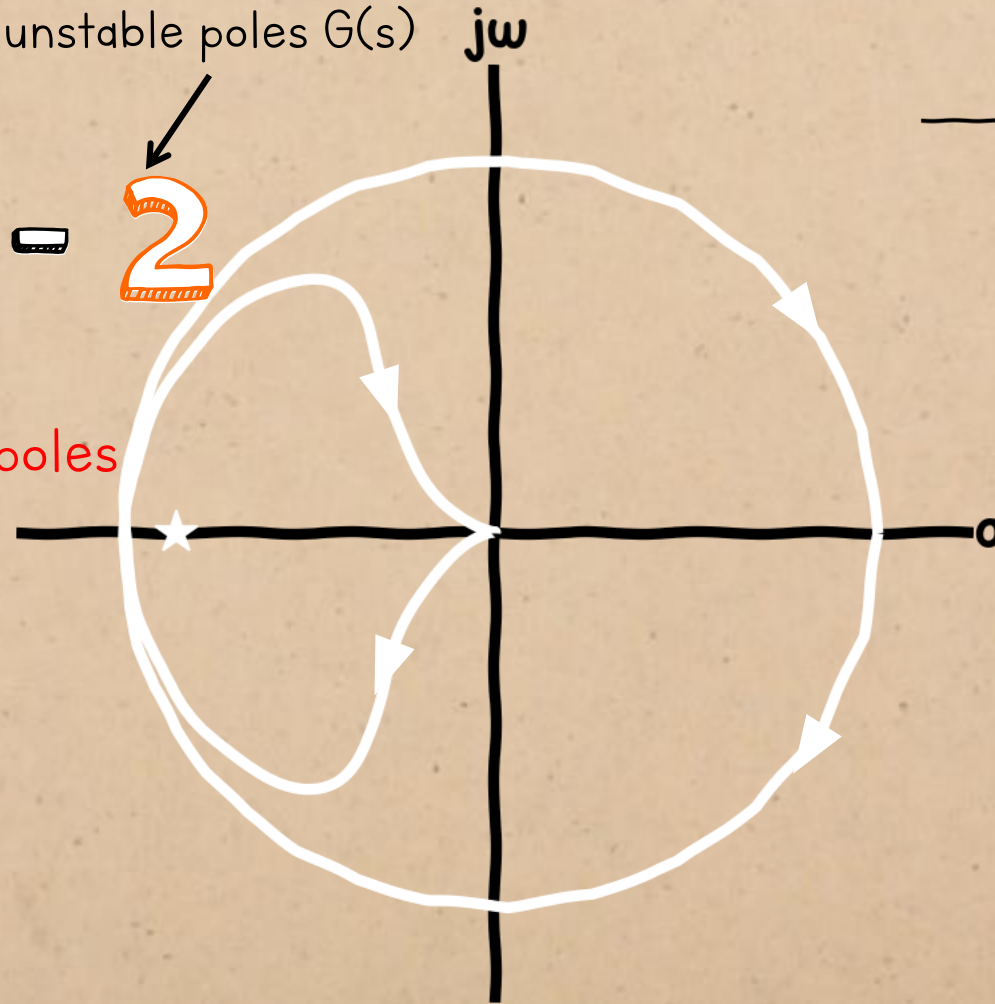
FREQUENCY DOMAIN

clockwise encirclements of -1

$$-2 = 0 - 2$$

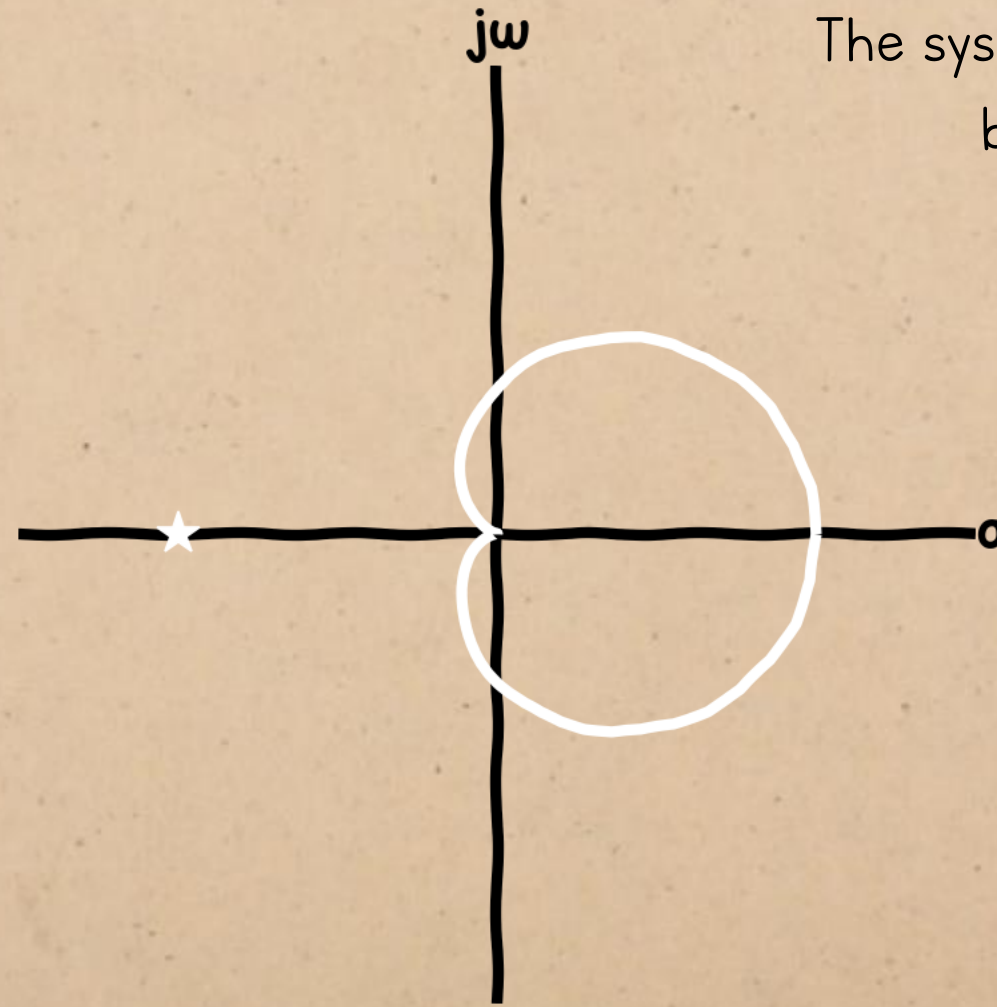
= unstable closed-loop poles

unstable poles $G(s)$



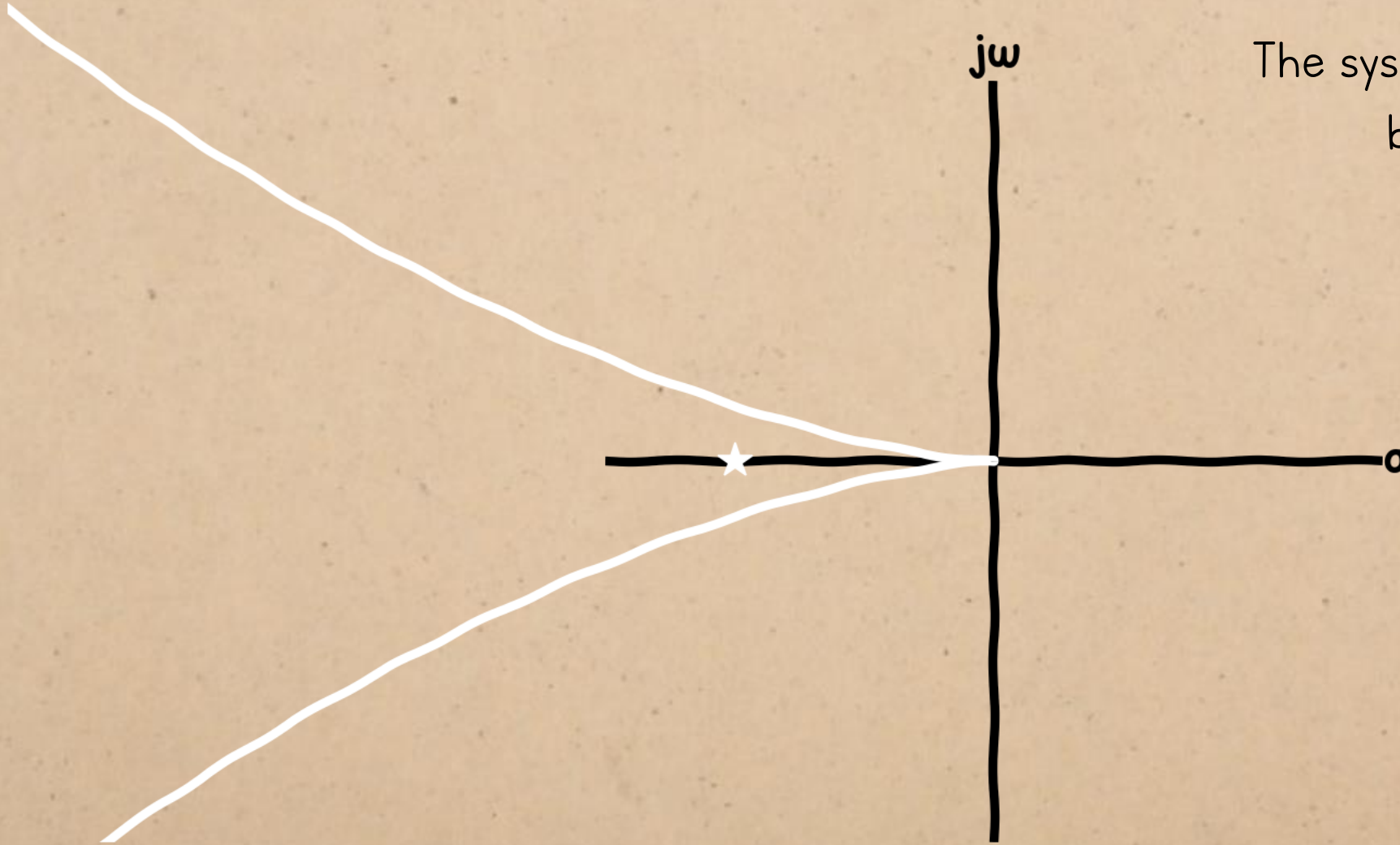
But this system has two unstable poles in open loop, so we need at least two anticlockwise encirclements to get no zeros.

FREQUENCY DOMAIN



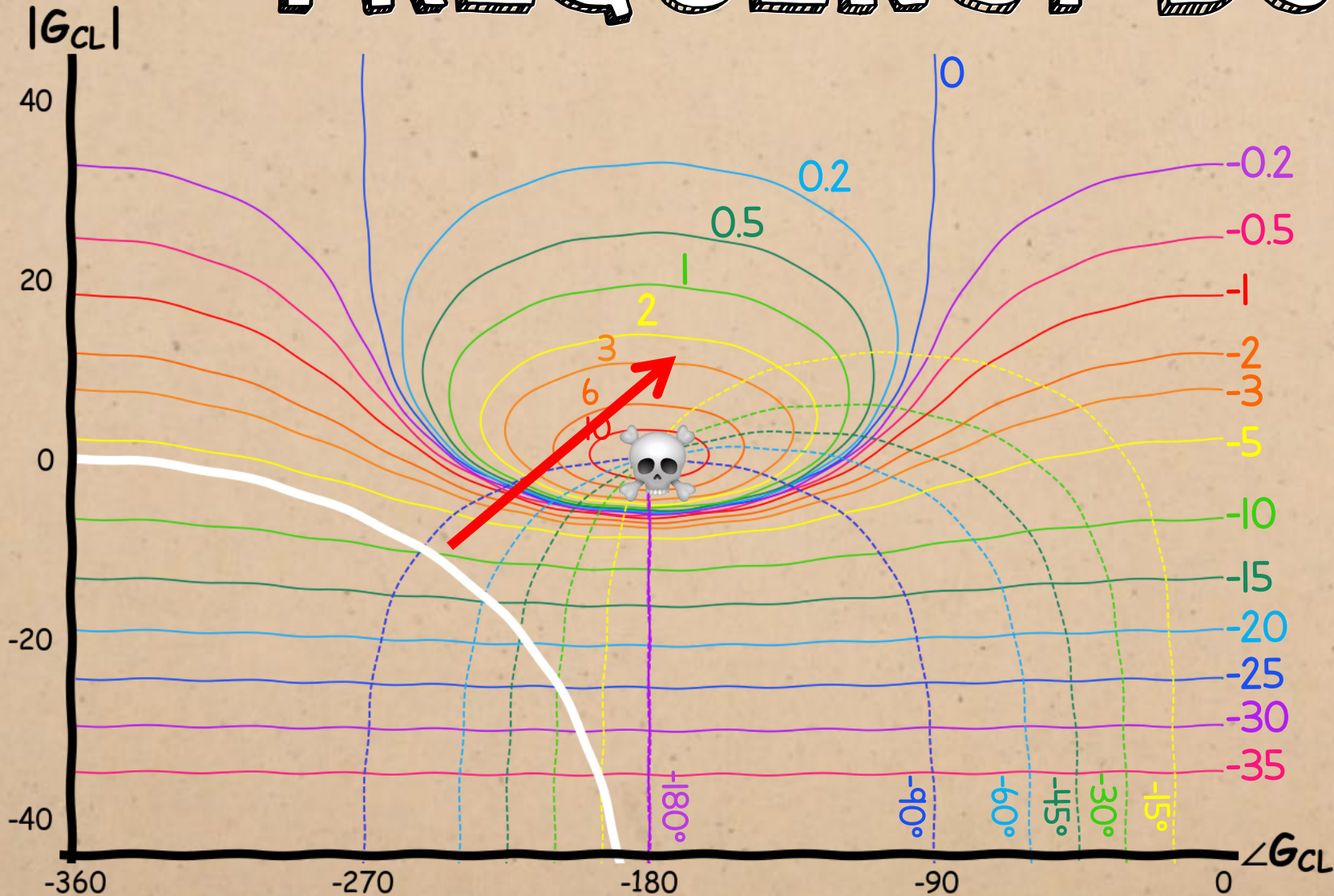
The system can't be stabilized just by changing the gain.

FREQUENCY DOMAIN



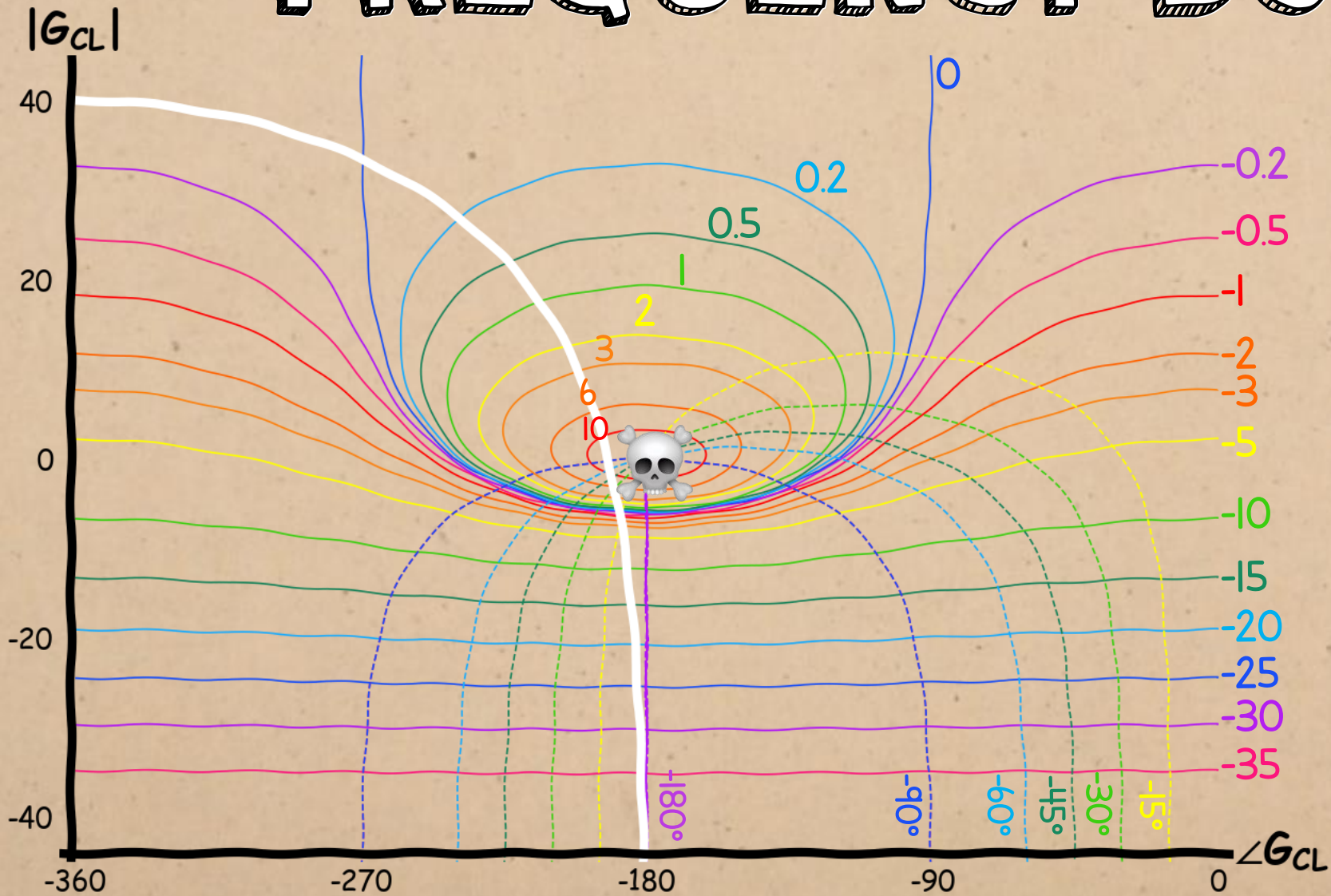
The system can't be stabilized just by changing the gain.

FREQUENCY DOMAIN



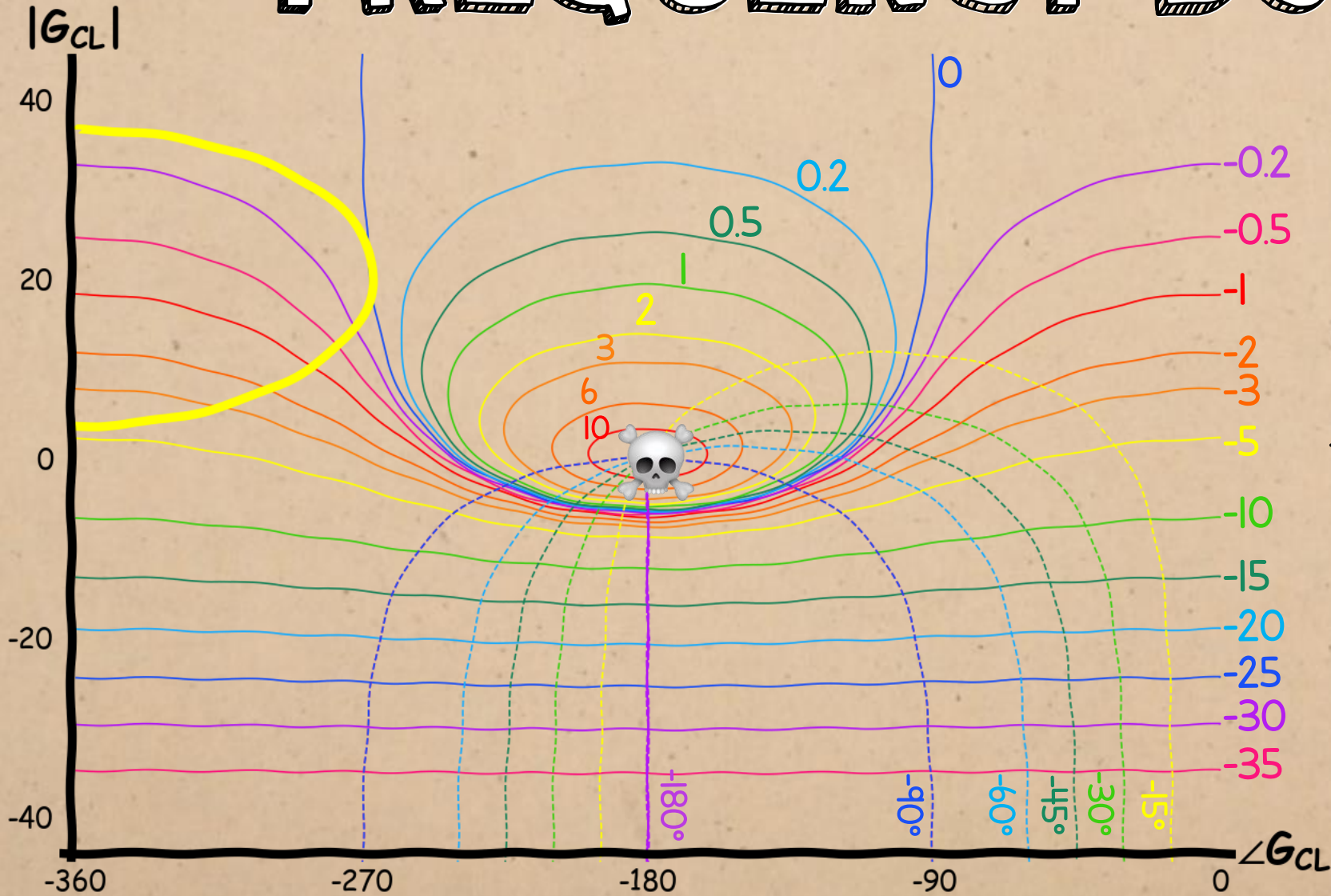
To stabilize the system, we need the response to pass the -1 point.

FREQUENCY DOMAIN



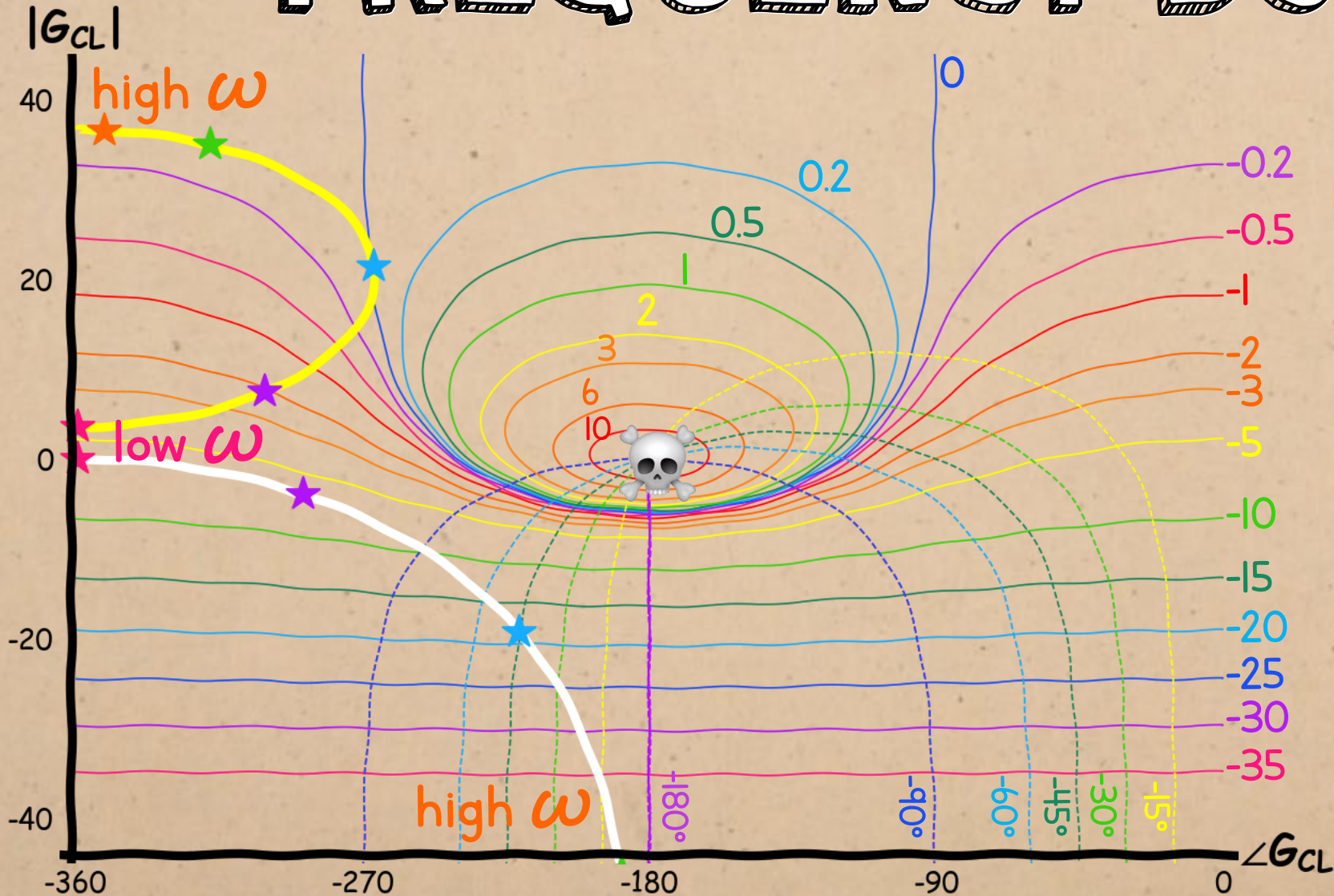
More gain won't help.

FREQUENCY DOMAIN



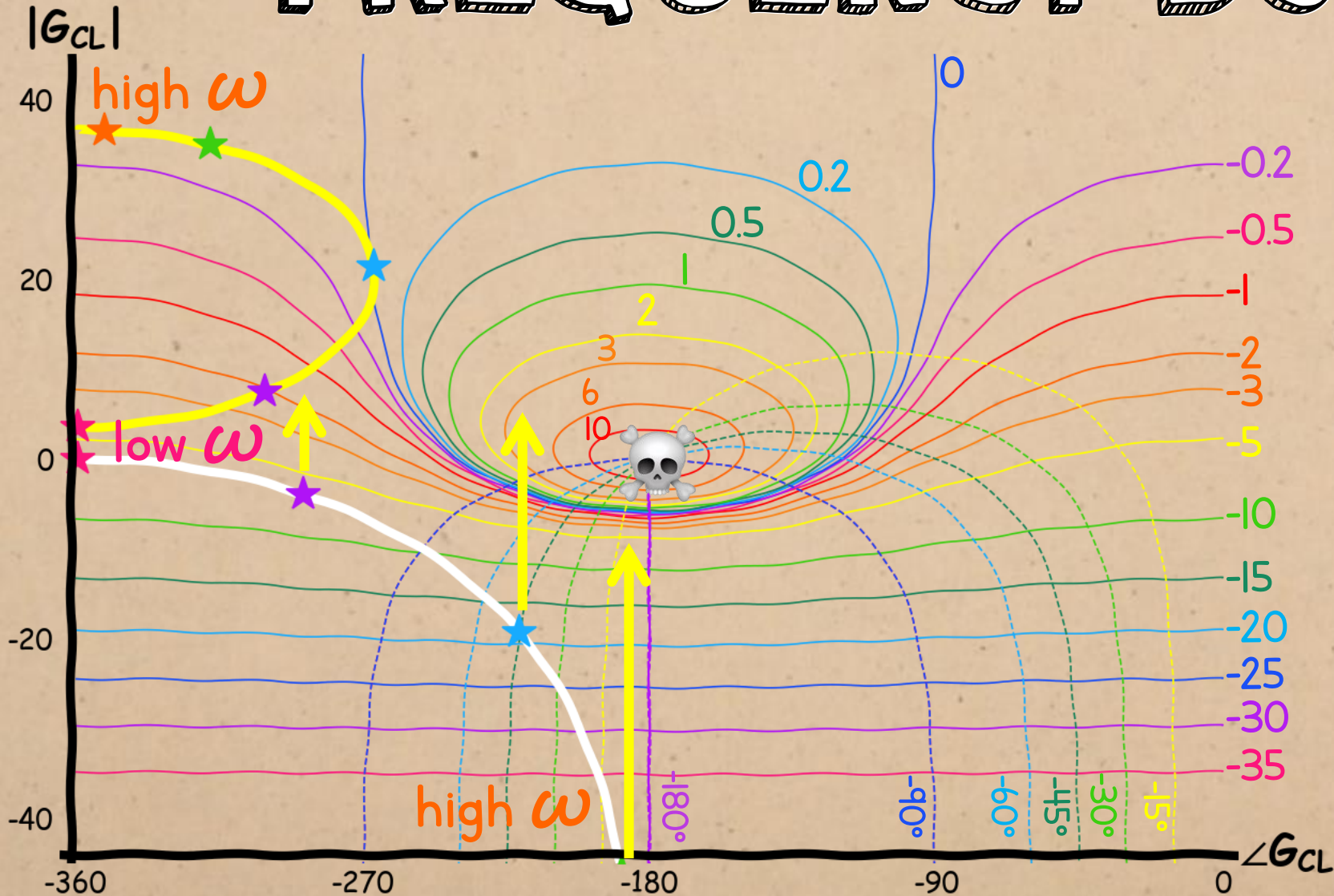
This is the response of the compensator we came up with before.

FREQUENCY DOMAIN



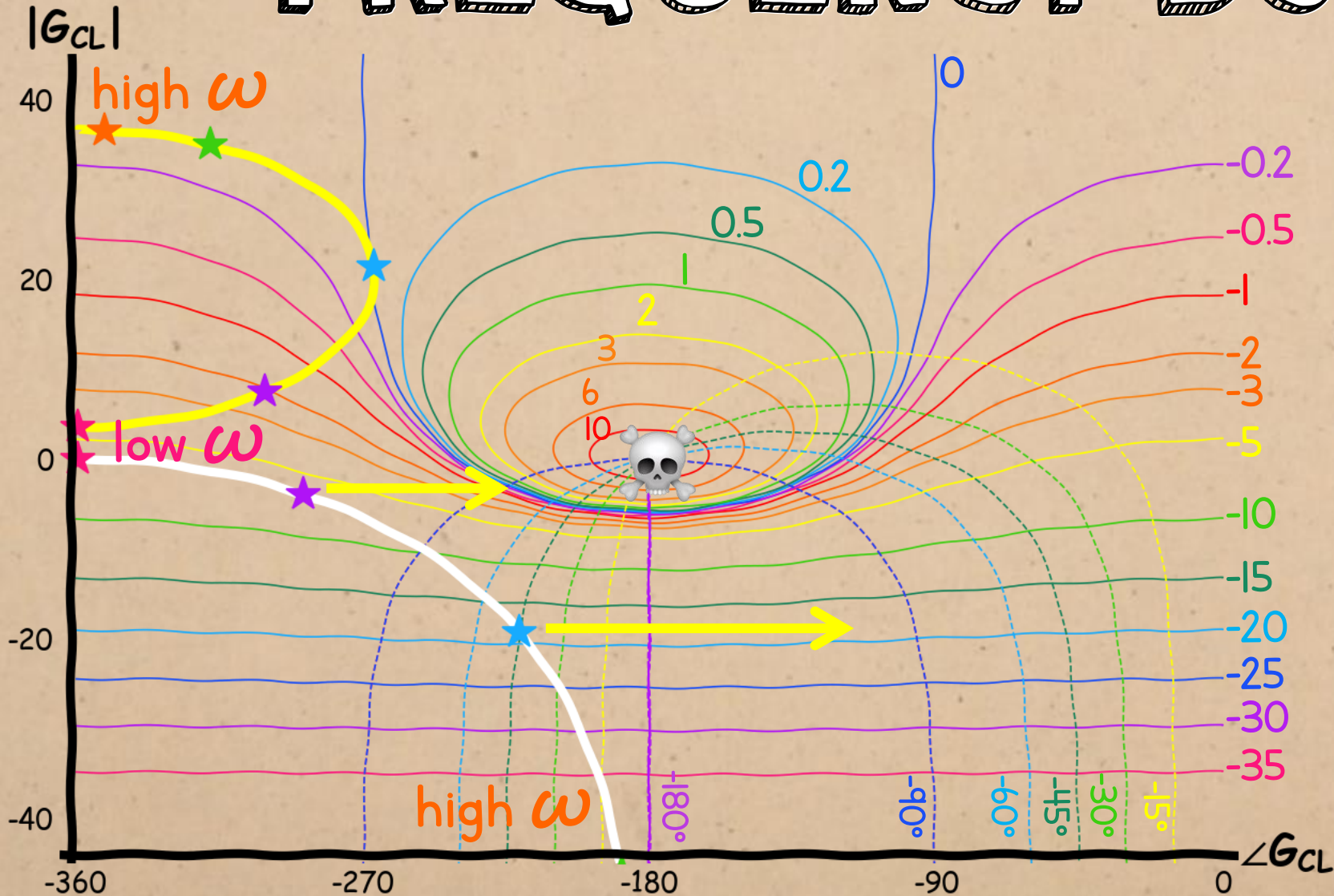
If we compare the same frequencies for the compensator and plant...

FREQUENCY DOMAIN



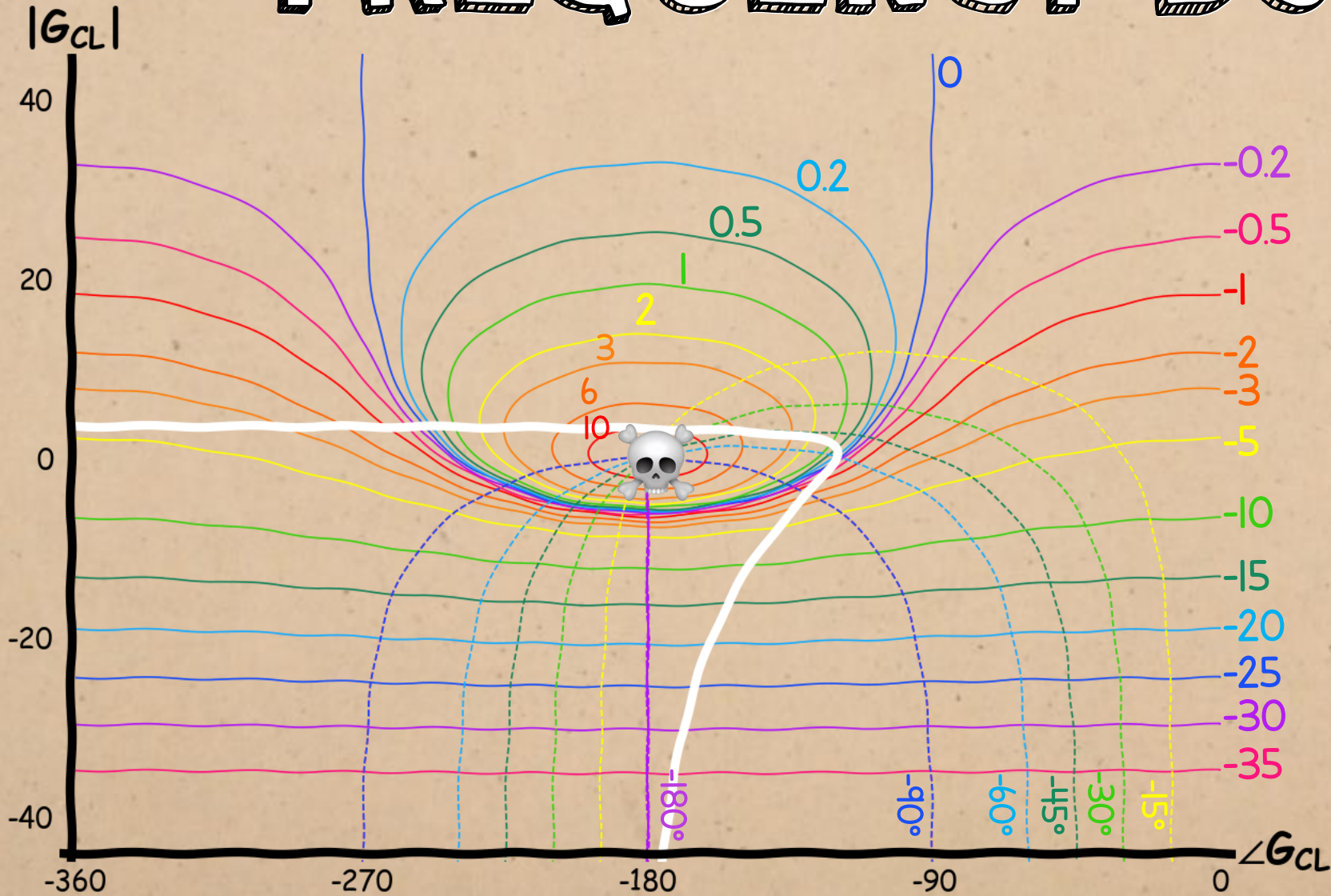
The compensator adds more gain as the frequency increases,

FREQUENCY DOMAIN



And it primarily adds phase in the middle range.

FREQUENCY DOMAIN



So when we combine them, we get the -1 crossing we need.

COMPENSATORS

And that's all there is to compensation:
You're just adding zeros and poles (that is
adding, or subtracting phase), and then
choosing an appropriate gain.



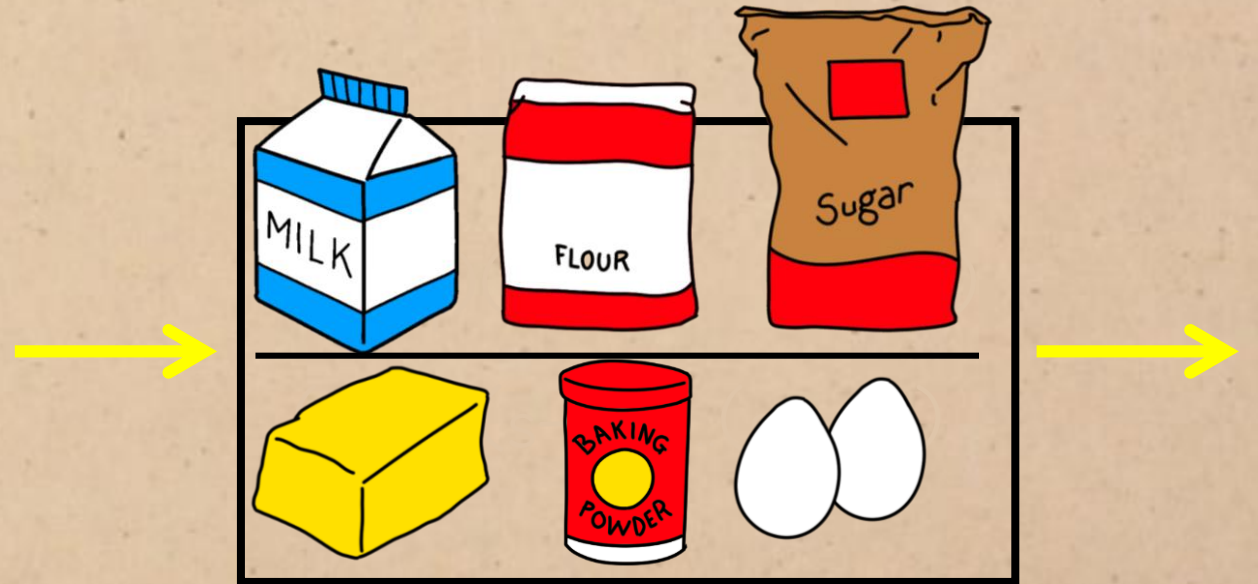
$$\frac{K(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)}$$

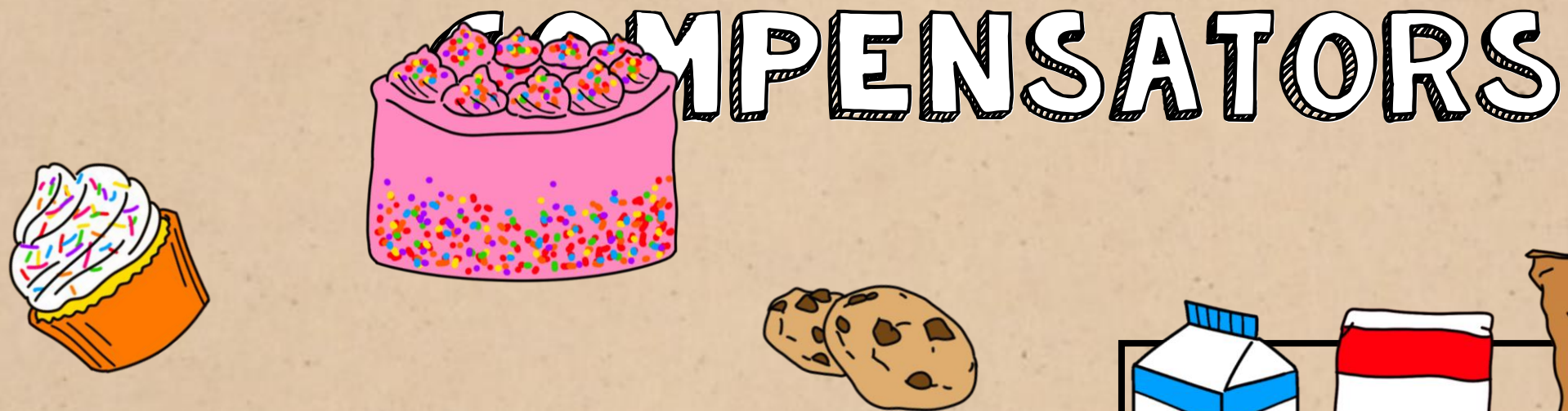


But it's not always easy to know where to
start if you're thinking in terms of a general
controller transfer function like this.

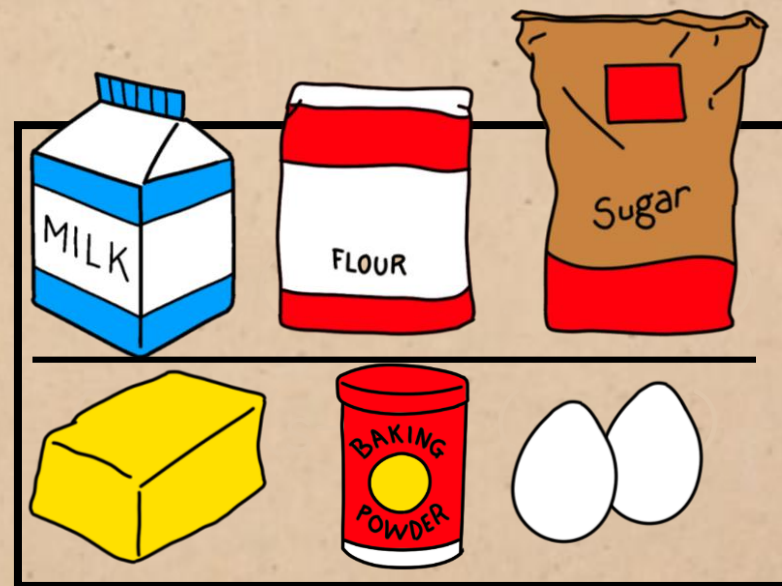
COMPENSATORS

If zeros, poles and gain are
our basic ingredients...





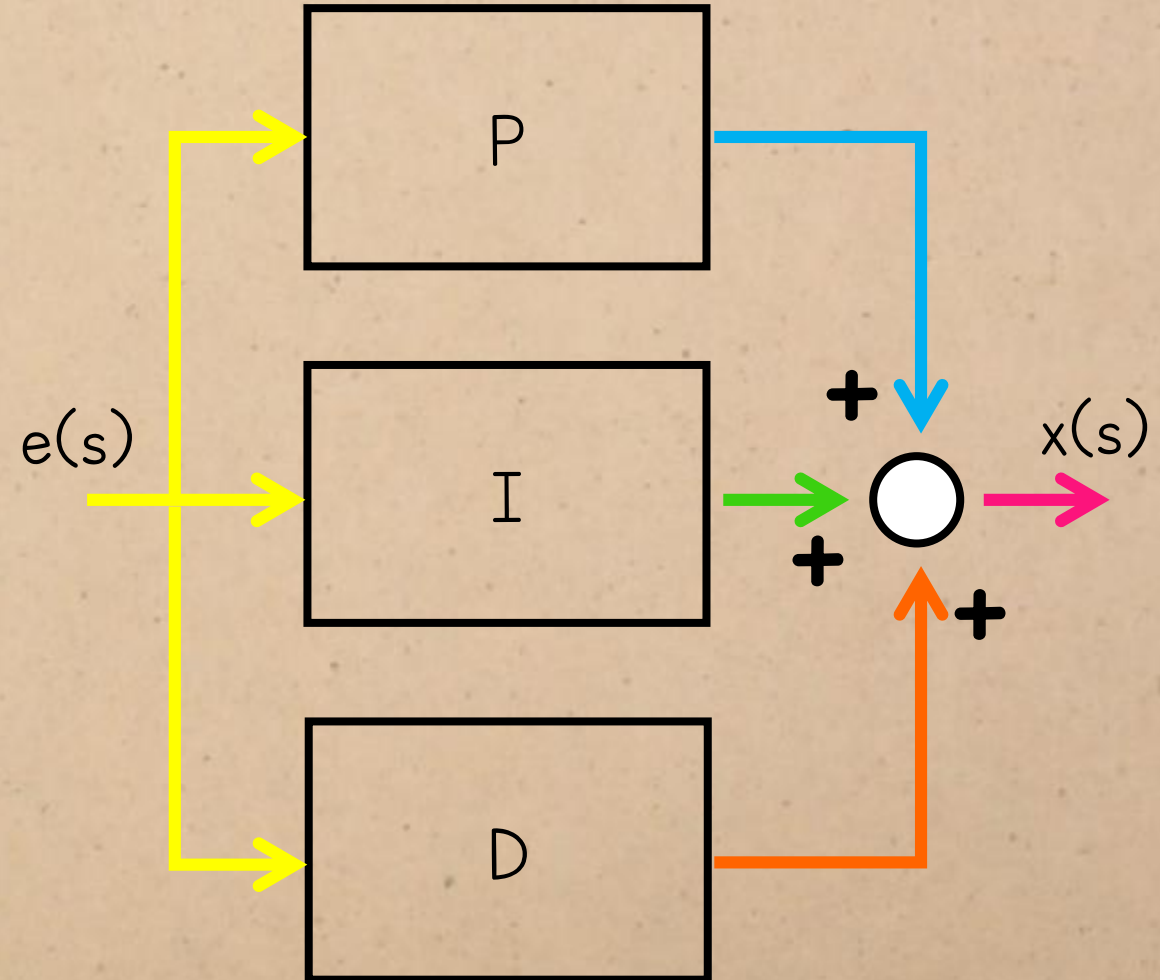
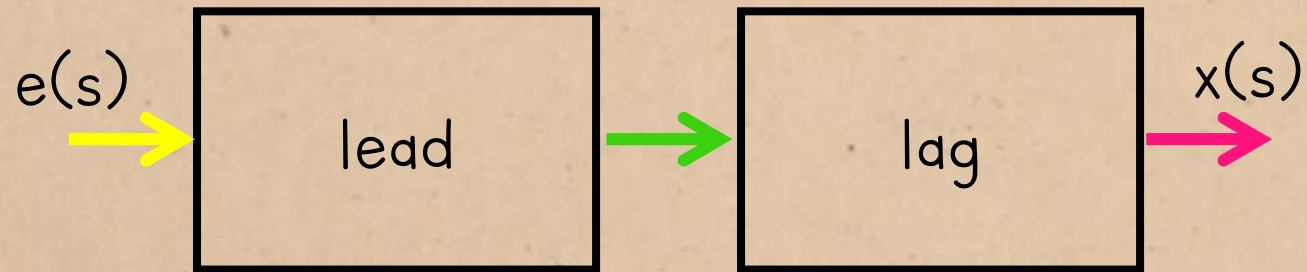
it might be helpful to learn some
tried-and-tested recipes to suggest
how we might combine them to get
certain results.



These compensator 'recipes' are what we're going
to learn for the rest of the semester.

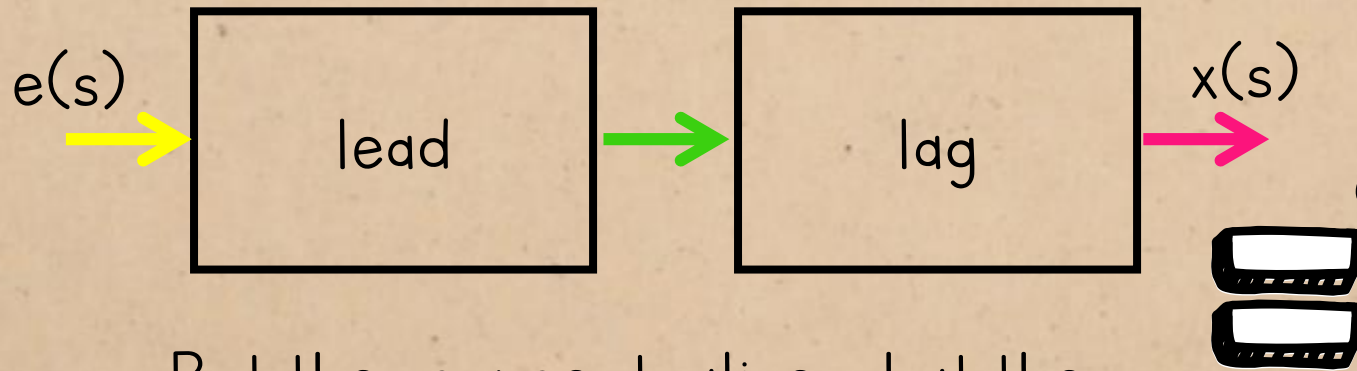
COMPENSATORS

The first two compensator types we'll cover are lead-lag and PID controllers

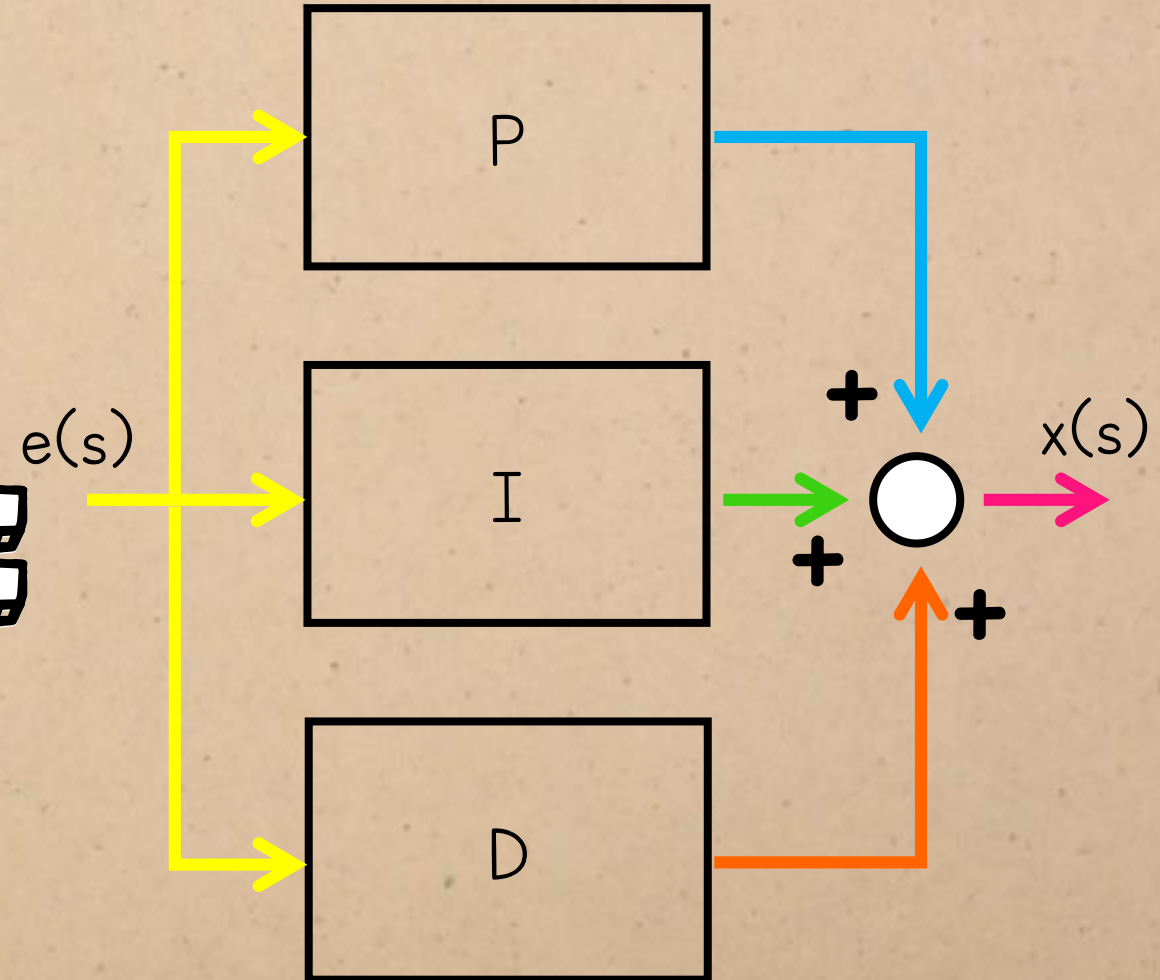


COMPENSATORS

Mathematically, these are the same
(i.e. they have the same transfer functions)

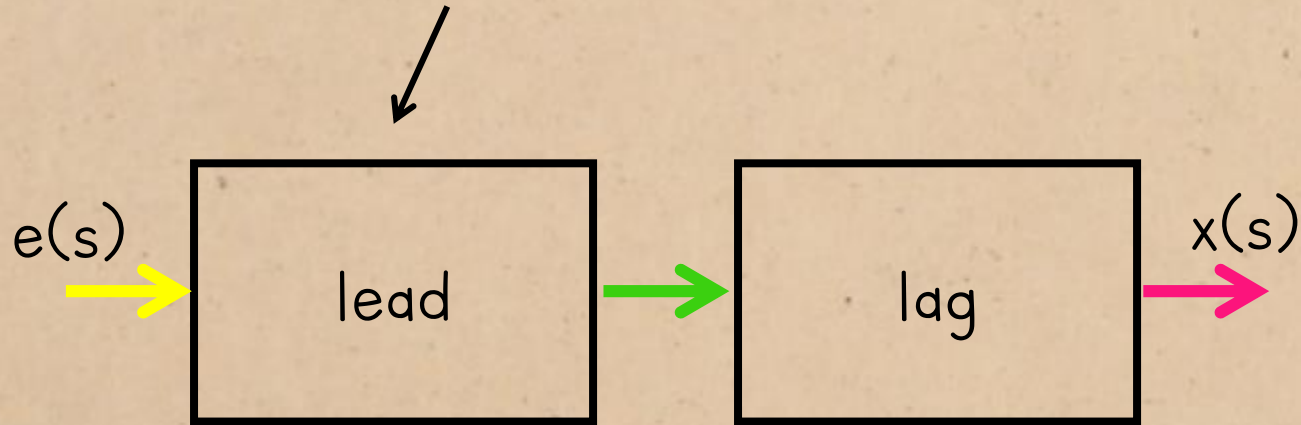


But they conceptualize what the
controller is doing in two different ways.



COMPENSATORS

Lead compensators create a
positive phase shift

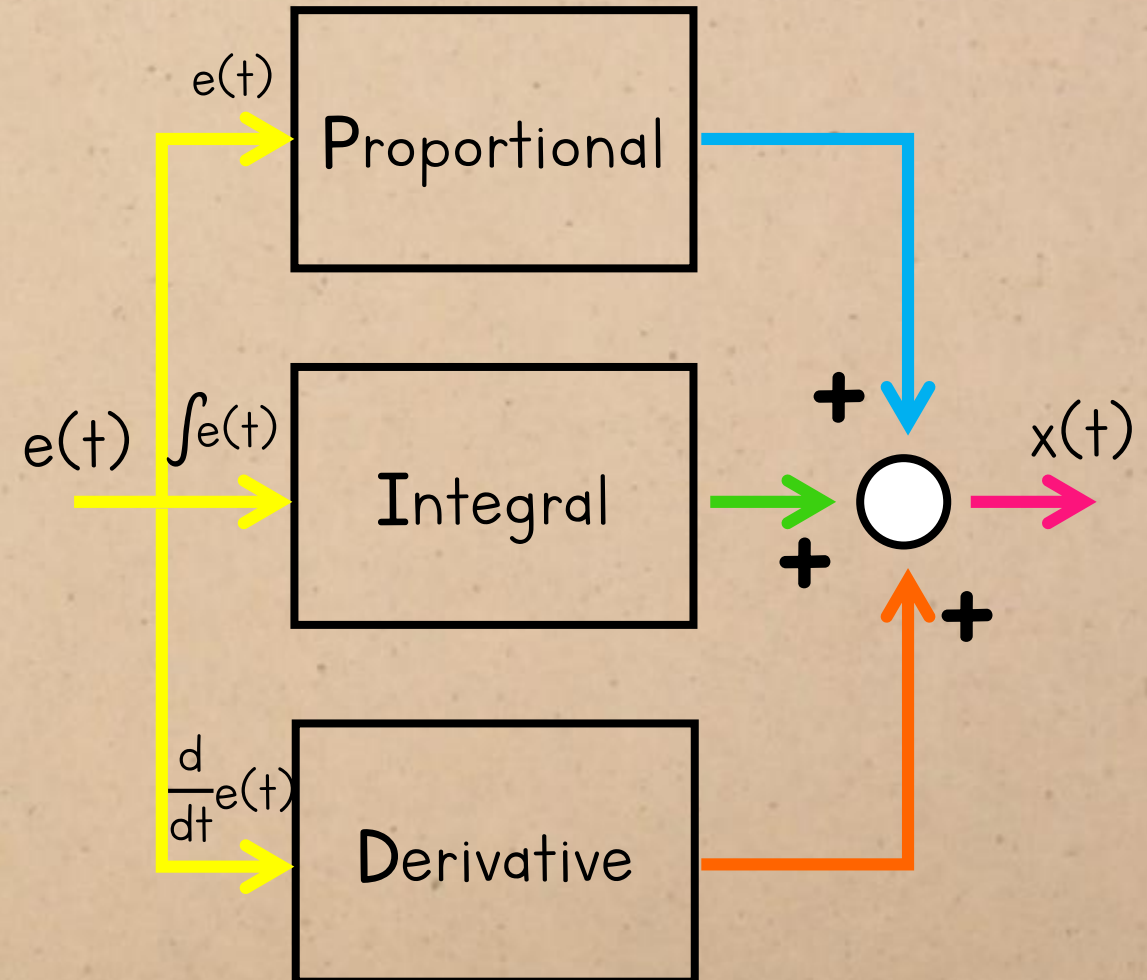


Lag compensators create a
negative phase shift

Lead-lag compensation works in terms of the **gain and phase** that the controller adds (or subtracts) from the plant.

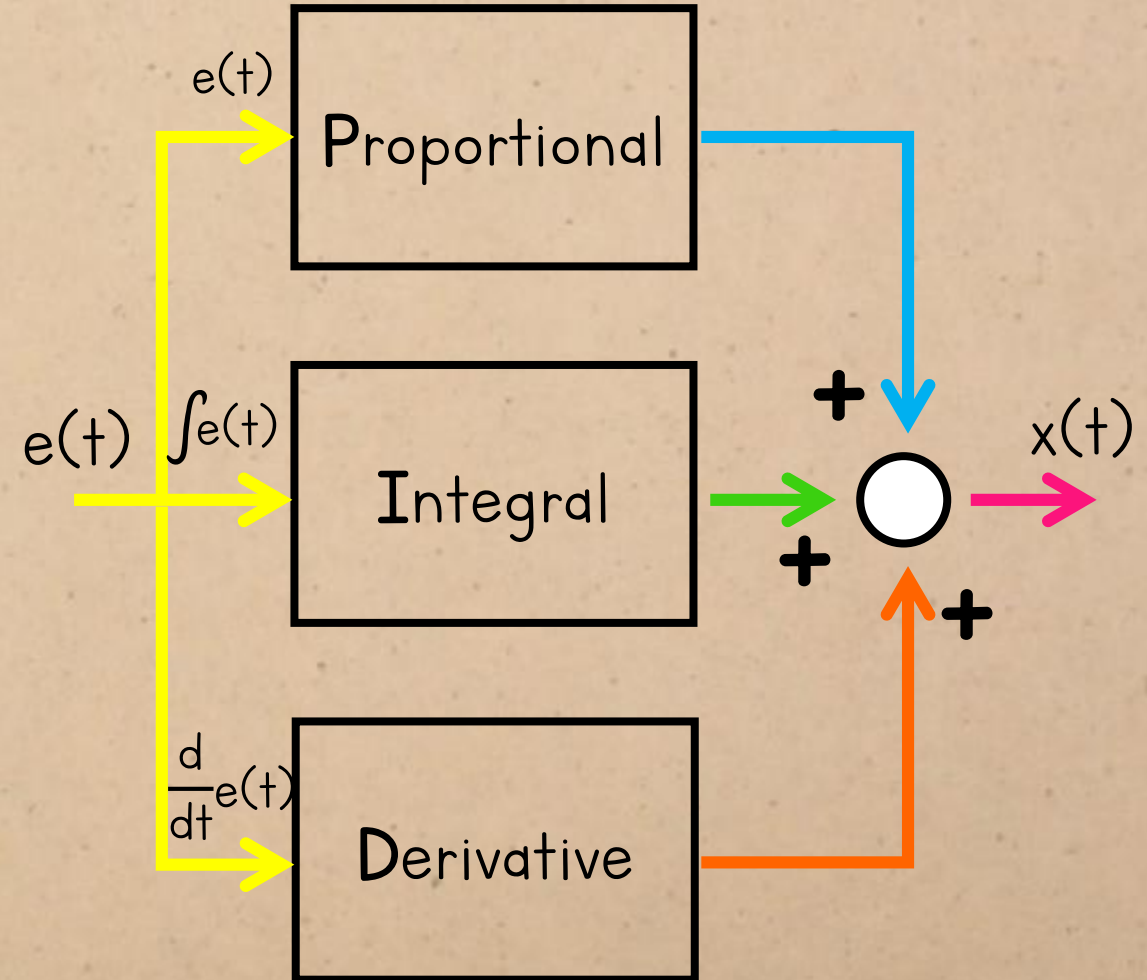
COMPENSATORS

With PID control, we think of the controller's output as a weighted combination of the **error**, its **integral** and its **derivative**.



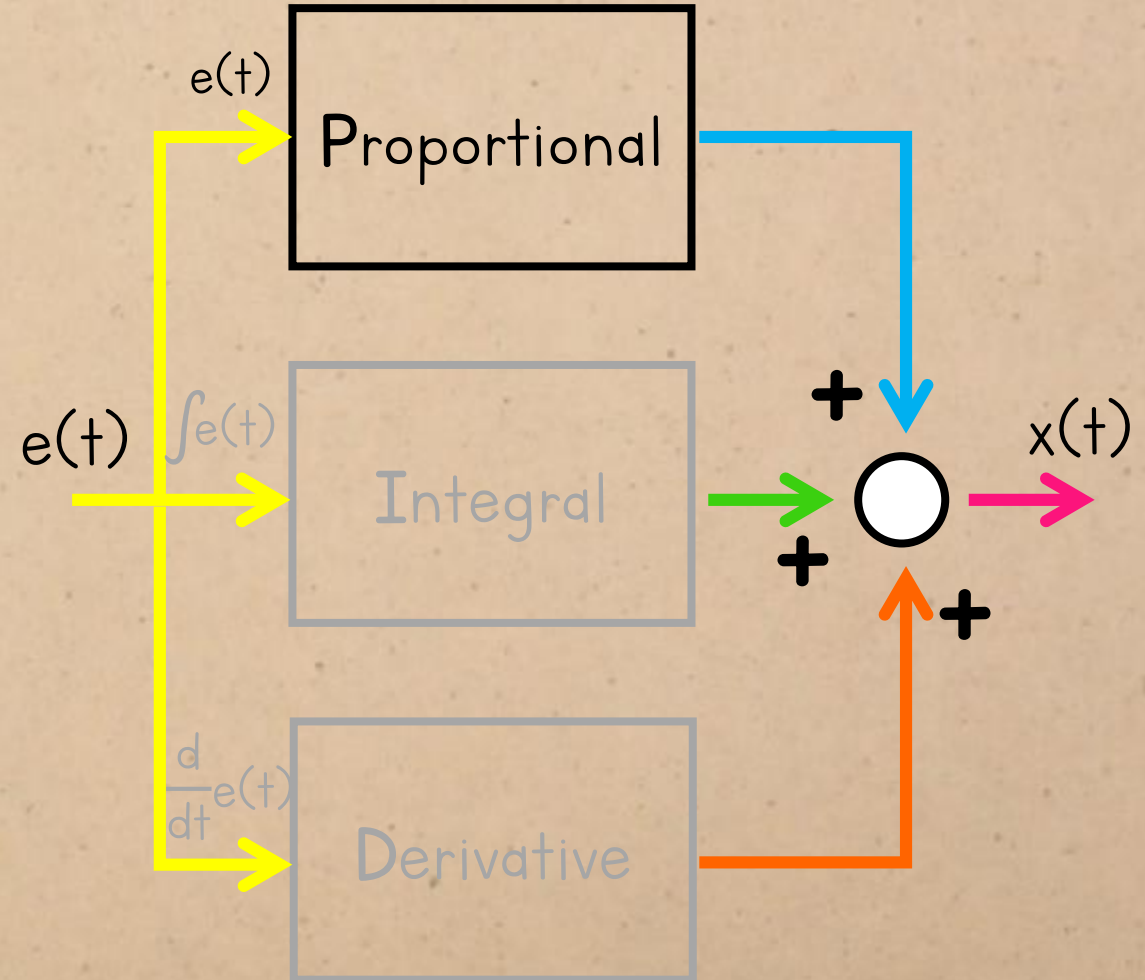
COMPENSATORS

We don't need to include all three elements at once.



COMPENSATORS

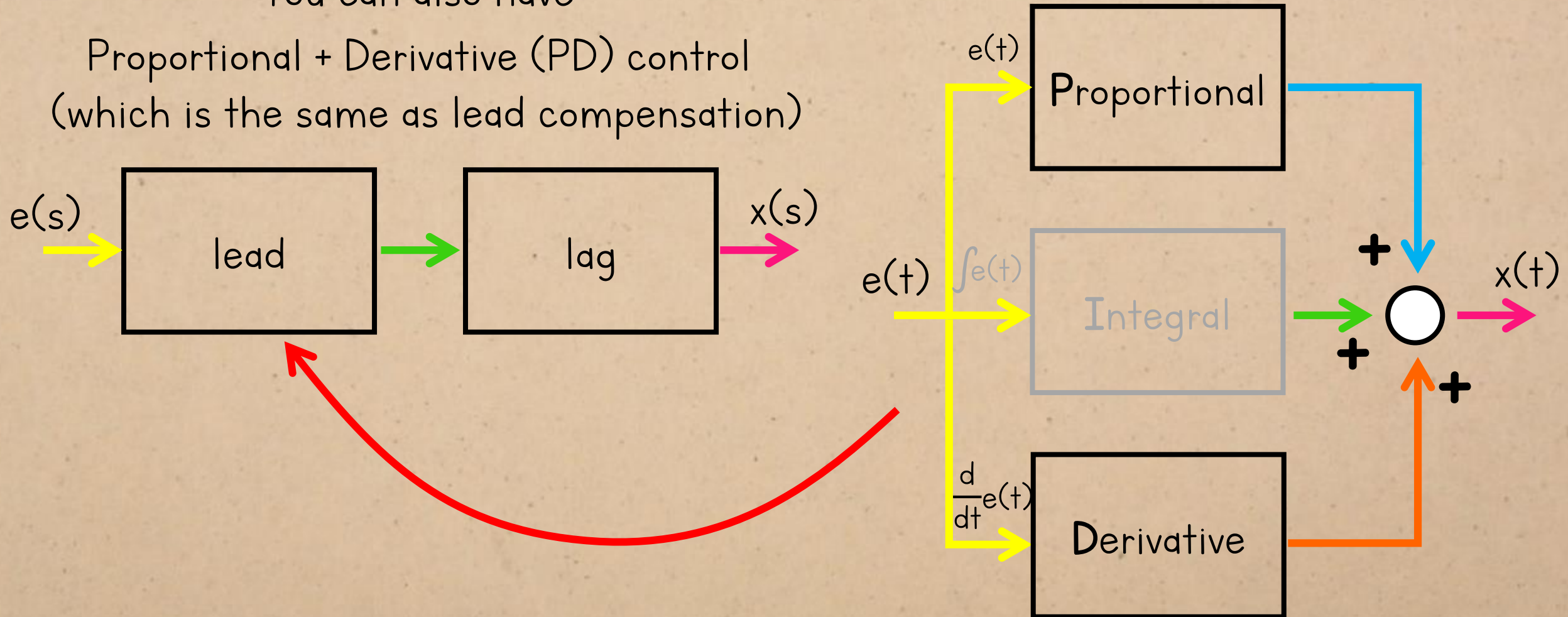
Proportional control alone is effective if the root locus already reaches satisfactory pole locations.



COMPENSATORS

You can also have

Proportional + Derivative (PD) control
(which is the same as lead compensation)



COMPENSATORS

And also

Proportional + Integral (PI) control
(which is a special case of lag compensation)

